



# Teacher judgments of student self-regulated learning – Examining the effect of student and teacher characteristics

Sabrina Nadja Brunner<sup>1</sup> · Yves Karlen<sup>1</sup>

Received: 29 October 2025 / Accepted: 15 April 2026  
© The Author(s) 2026

## Abstract

Students skilled in self-regulated learning (SRL) can plan, monitor, and reflect on their learning, using strategy knowledge (SK) to select and apply strategies. Teachers often struggle to assess SRL and rely on non-diagnostic cues, particularly when their knowledge of SRL is limited. Biased evaluations could affect how teachers support students and impact their performance. This study investigated teachers' accuracy in judging students' SK across five SRL components (cognition, metacognition, time management, management of the learning environment, and motivation regulation) by comparing teachers' SK judgments with students' SK test scores. Specifically, we explored judgment biases related to students' academic achievement, gender, and socioeconomic status. Further, teacher characteristics that may contribute to biased judgments were examined (knowledge of SRL, self-concept, experience, and gender stereotypes). The sample included 200 students and 69 teachers. Teachers' relative judgment accuracy was assessed using correlations with cluster-robust standard errors. Multilevel models were used to analyze how student characteristics affected teachers' SK judgments. Cross-level interactions were employed to investigate the impact of teacher variables on judgment biases. Results indicated low accuracy and component-specificity of teachers' judgments. Furthermore, their judgments were affected by achievement and gender biases. These may be attributed to a halo effect, in which teachers' overall impression of a student influenced their SK judgments. Teachers' knowledge of SRL, self-concept, and experience were not associated with reduced biases, but were (trend-wise) associated with stronger biases, albeit not consistently across components. These findings question teachers' ability to distinguish between students' SRL skills and their overall impressions of students.

**Keywords** Self-regulated learning · Teacher judgment accuracy · Metacognitive strategy knowledge · Judgment bias · Multilevel models

---

✉ Sabrina Nadja Brunner  
sabinanadja.brunner@ife.uzh.ch

<sup>1</sup> University of Zurich, Kantonsschulstrasse 3, Zurich 8001, Switzerland

## Introduction

The ability to self-regulate one's own learning is a key educational goal (OECD, 2019). To prepare for their future careers in a world where professions are rapidly evolving, accelerated by the fast-paced evolution of AI, students must develop their ability to monitor and regulate their learning (Kirschner & Stoyanov, 2020). This ability to plan, monitor, and reflect on their learning intentionally is referred to as self-regulated learning (SRL) (Zimmerman, 2000). Despite its importance, students often struggle with SRL (Jeong and Feldon, 2023; Karlen, 2016a). Using SRL strategies increases students' cognitive load during learning and requires mental effort, which must be allocated between processing the learning content and regulating the learning process metacognitively (de Bruin et al., 2020). Furthermore, some SRL strategies are task-specific and not equally useful across learning contexts (Greene et al., 2015), underscoring the importance of students' strategy knowledge (SK) regarding when and how to apply strategies (Flavell, 1979). To support students' SRL development, adaptive teaching is crucial: teachers are expected to guide students through their learning process, based on their individual strengths and weaknesses in SRL (Karlen & Hertel, 2024), and to offer tailored support depending on each student's needs (Corno, 2008). However, teachers have shown difficulty in judging their students' learning skills (Urhahne & Wijnia, 2021) and have been found to make biased judgments in SRL-related aspects such as motivation (Urhahne et al., 2013) and engagement (Kaiser et al., 2013). The current study extended this research by focusing on teachers' judgments of students' SRL, specifically students' SK. It examined how accurately teachers could judge their students' SK, whether their judgments were systematically biased by students' characteristics, and whether these biases were related to teachers' knowledge of SRL, self-concept, experience, and gender stereotypes. Since teachers' judgments of their students' SRL could affect their adaptive teaching (Corno, 2008) and their track recommendations (Sneyers et al., 2019), unbiased judgments are of great importance.

## Theory

### Self-regulated learning

SRL is relevant to the entire learning process and refers to the ability to manage it independently and purposefully (Zimmerman, 2002). There are several definitions and models of SRL, but at its core, it includes cognitive, metacognitive, and motivational components of learning (Boekaerts, 1997; Schraw et al., 2006). The cognitive component includes strategies that facilitate information processing, such as rehearsal, elaboration, and organization (Boekaerts, 1997). The metacognitive components include strategies that support planning, monitoring, and reflection on the learning process, for example, by monitoring learning gains and evaluating goal attainment after completing a task (Schraw et al., 2006). The motivational component encompasses strategies to regulate (and often increase) one's motivation, such as turning boring tasks into a game or employing positive self-talk (Miele et al., 2024). A further important aspect of SRL is the adaptive management of resources (Waldeyer et al., 2020), with motivation regulation sometimes considered part of resource management, when motivation is conceptualized as a resource (Dresel et al., 2015). Further-

more, relevant resources include the learning environment and learning time, which must be managed efficiently (Dresel et al., 2015). Overall, the SRL process is complex. It involves activating and coordinating cognitive, metacognitive, motivational, and resource-oriented strategies to achieve desired goals (Boekaerts, 1997).

To use strategies effectively and adaptively, learners need knowledge of these strategies (Dresel et al., 2015; Flavell, 1979; Miele et al., 2024). SK refers to the knowledge of when and how to use specific strategies (Flavell, 1979). SK is closely intertwined with knowledge of tasks and their requirements, as well as knowledge of oneself as a learner, all of which fall under the broader category of metacognitive knowledge (Flavell, 1979). SK, strategy use, and metacognitive experiences interact (Efklides & Schwartz, 2024) and collectively influence students' learning success, thereby affecting their academic achievement (de Boer et al., 2018; Richardson et al., 2012).

### Assessment of self-regulated learning

Teachers' competencies play an important role in explaining their SRL-related practices (Karlen et al., 2020). In addition to knowledge of SRL, as well as the beliefs and motivations that support its promotion (Karlen, Hirt et al., 2023), teachers are expected to monitor and assess their students' SRL, with accurate judgments of students' current SRL as a prerequisite for adaptive support (Corno, 2008). The importance of teachers' judgments is further underscored by findings showing their impact on teachers' behavior. Teachers tend to display more positive interactions with students they perceive as having higher potential (Harris & Rosenthal, 1985). Additionally, they are inclined to provide them with more instructional input, more challenging material, and more opportunities to participate (Harris & Rosenthal, 1985), which may contribute to a self-fulfilling prophecy affecting students' development (H. M. Cooper, 1979). Thus, especially teachers' underestimation of students' performance could harm later student achievement (Kolovou et al., 2024; Zhou & Urhahne, 2013) by undermining their self-efficacy (Won et al., 2017) and self-concept (Zhu et al., 2018). In contrast, teachers' overestimation might even foster students' further development (Bergold & Steinmayr, 2023). However, since teachers consider their judgments of students' learning behavior when making recommendations for students' educational trajectories, such as track placement (Sneyers et al., 2019), accurate SRL judgments are essential.

Admittedly, forming judgments is a highly complex process (Funder, 1995; Loibl et al., 2020). Given the multitude of available information, teachers need to know which cues they rely on (Tannert et al., 2026; Koriati, 1997; Loibl et al., 2020). Previous research has found that teachers often rely on surface cues, such as off-task behavior, to judge SRL (Callan & Shim, 2019; Dignath & Sprenger, 2020), even though such cues are not truly diagnostic and may stem from various causes. Additionally, teachers mainly mentioned learning dialogue or observations as ways to assess their students' SRL, and few were aware of diagnostic activities, such as specific SRL tasks or think-aloud (Bäuerlein et al., 2023; Michalsky, 2017), suggesting limited knowledge of SRL assessment.

When teachers struggle to identify relevant cues, they are more likely to base their judgments on global impressions or salient student characteristics, known as the Halo effect (W. H. Cooper, 1981). A positive Halo effect implies a higher judgment depending on a positive overall impression, while a reverse Halo effect implies a lower judgment due to a negative overall impression or an unrelated but negatively judged characteristic (Noor

et al., 2023). This closely aligns with confirmation bias, which describes the tendency for people to seek out information that supports their initial beliefs and to interpret information received in ways consistent with those beliefs (Nickerson, 1998). In line with this, teachers questioned standardized test results that were inconsistent with their prior assumptions and sought alternative explanations, without indicating the intention to question their previous beliefs (Vanlommel et al., 2017).

Beyond theoretical considerations, empirical evidence indicates that Halo effects affect teachers' judgments of students' academic achievement (Jansen et al., 2021; Schmidt et al., 2023), motivation (Urhahne et al., 2013), and engagement (Kaiser et al., 2013). To our knowledge, no previous study has examined these effects regarding teachers' SRL judgments. However, judging students' study skills appears particularly challenging for teachers compared to judging academic achievement (Urhahne & Wijnia, 2021). Furthermore, teachers' SRL judgments may be particularly susceptible to bias due to the low observability of diagnostic cues and teachers' limited knowledge of SRL (Tannert et al., 2026).

Building on these challenges, assessing SK rather than students' actual strategy use introduces an additional layer of complexity. An absence of students' use of strategies does not necessarily indicate low SK and may also be due to low motivation or prioritizing extracurricular goals (Wesarg-Menzel et al., 2023). This discrepancy between demonstrated SRL and underlying SK poses an additional challenge for teachers' SK judgments, aligning with the low accuracy of teachers' SK judgments in our previous study (Karlen, B uerlein et al., 2023). Nevertheless, teachers' ability to accurately assess students' SK and distinguish between students' strategy use and their underlying SK is essential for promoting adaptive SRL (Karlen & Hertel, 2024). Whether a student lacks SK or has difficulty implementing it into practice is an important differentiation, with distinct promotion approaches. SRL can be promoted directly through strategy instruction or by offering an SRL-supportive learning environment that allows students to practice their SRL skills (Perry, 2013). To correct potential erroneous beliefs (Onan et al., 2024) and enhance SK, adaptive strategy instruction is highly important, whereas the provision of an SRL-supportive learning environment is insufficient (Karlen 2016b).

## Effects of student characteristics on teachers' SK judgments

When teachers have inadequate cues or are unaware of relevant indicators for accurately diagnosing their students' SK, they are more likely to rely on their overall impression or on other salient student characteristics (W. H. Cooper, 1981). In this case, teachers' judgments are not based solely on students' SK but also on other student characteristics. If SK is systematically over- or underestimated based on students' characteristics, such as academic achievement, gender, or socioeconomic status (SES), such bias could perpetuate or deepen existing educational inequalities.

Previous research on the effect of student characteristics on teachers' judgments showed a systematic underestimation of the SK for students with lower academic achievement (Carr & Kurtz, 1991). Regarding gender, teachers systematically underestimated boys' planning abilities and learning motivation (Sneyers et al., 2020). However, this study relied on self-report measures, and the authors suggest that the observed bias may stem from boys' tendency to overestimate their performance, whereas girls tend to underestimate theirs (Cole et al., 1999). No gender effect on teachers' judgments of students' SK in math and reading was

observed (Carr & Kurtz, 1991). In addition to these unclear gender effects, teachers underestimated students' motivation to learn among those from low SES backgrounds (Sneyers et al., 2020). This may stem from the association between SES and behavioral regulation (Størksen et al., 2015) or school grades (Tan, 2024), both of which teachers reported relying on when diagnosing SRL (Callan & Shim, 2019; Dignath & Sprenger, 2020). Even in a laboratory setting, where case vignettes manipulated students' SES and social background, teachers judged the willingness to put in effort as lower for low-SES students than for high-SES students (Tobisch & Dresel, 2017). Overall, empirical research on teachers' judgment accuracy regarding SRL remains sparse and typically addresses only a limited number of SRL components (Urhahne & Wijnia, 2021). Even fewer investigations examined teachers' judgment of students' SK. More research is needed to understand the interplay between student characteristics and their impact on teachers' judgments. Previous research indicates that higher achievement, female gender, and higher SES are associated with higher SK (Brunner et al., 2025). Accordingly, biased judgments can be inferred only when the correlation between teachers' judgments and these characteristics exceeds the actual correlation (W. H. Cooper, 1981). Such a bias could result from teachers' overestimations of these essentially true associations.

### Effects of teacher characteristics on their SK judgments

In addition to students' characteristics, teachers' own characteristics may also affect their judgments (Loibl et al., 2020; Südkamp et al., 2012). Differences in teachers' knowledge, beliefs, and motivation could impact their judgment accuracy (Funder, 1995). Regarding knowledge, teachers with a better understanding of SRL tended to rely less on non-diagnostic cues when assessing students' SRL (Dignath & Sprenger, 2020). In addition, teachers' general diagnostic knowledge (e.g., how to proceed in forming a systematic diagnosis) has been associated with higher teacher competence in judging learning behavior (Klug et al., 2016). However, this knowledge did not always translate into teachers' assessment practices (Michalsky, 2017). Neither teachers' understanding of the SRL concept (content knowledge) nor their knowledge of diagnostic cues and assessment methods to diagnose SRL (assessment knowledge) predicted judgment accuracy (Karlen, Bäuerlein et al., 2023). Thus, findings regarding the impact of knowledge on teachers' judgment accuracy remain inconclusive. Nevertheless, systematic judgment biases, such as the halo effect, are expected to be stronger when raters have limited knowledge about the concept being evaluated (W. H. Cooper, 1981).

Closely aligned with knowledge (Paulick et al., 2016), teachers' self-concept reflects their perceived competence in a specific domain (Bong & Skaalvik, 2003). Teachers' self-concept has been shown to affect their teaching behavior (Yeung et al., 2014), but was not associated with their judgment accuracy (Karlen, Bäuerlein et al., 2023). In line with this, teachers' self-efficacy did not predict teachers' diagnostic competence (Klug et al., 2016). The authors interpreted this as teachers' poor self-assessment, since self-efficacy represents the self-perceived confidence in one's ability (Bong & Skaalvik, 2003). Accordingly, evidence regarding the effect of self-perceived competence on teachers' judgments is inconsistent. Regarding job experience, a systematic association with teachers' judgment accuracy was not observed, whereas familiarity with the assessed task was linked to higher accuracy

(Urhahne & Wijnia, 2021). Thus, while more years of teaching do not necessarily enhance accuracy, specific experience with the diagnosed concept appears beneficial.

Another aspect to consider is the role of stereotypes. Gender stereotypes have been repeatedly demonstrated among teachers, with considerable interindividual differences (Dersch et al., 2022; Retelsdorf et al., 2015). Teachers with stronger gender stereotypes were more likely to attribute students' failures in a stereotypical manner (Di Battista et al., 2024) and to assess students' reading abilities in line with gender-based expectations (Muntoni & Retelsdorf, 2018). Thus, gender stereotypes that attribute higher SRL abilities to females may contribute to potential gender biases in teachers' SRL judgments. However, to our knowledge, the investigation of gender stereotypes regarding SRL and their impact on teachers' judgments has not yet been conducted.

## The current study

Previous research has shown that teachers' judgment accuracy in assessing students' learning skills tends to be low (Urhahne & Wijnia, 2021). However, only a few studies have specifically examined teachers' accuracy in judging SRL. Existing studies have primarily relied on comparisons between teachers' ratings and students' self-reports (e.g., Friedrich et al., 2013; Sneyers et al., 2020), although students' self-reports are weakly related to SRL behavior (Dörrenbächer-Ulrich et al., 2021). Moreover, students' grades, gender, and SES appear to affect teachers' judgments of students' learning behavior (Carr & Kurtz, 1991; Sneyers et al., 2020). However, these student characteristics are interrelated (Caro et al., 2009; Verbree et al., 2023), and, to our knowledge, no prior study has included all of them in a single model to disentangle their distinct effects on teachers' SRL judgments. Additionally, little is known about interindividual differences in teachers' SRL assessment biases, although higher knowledge is typically associated with less judgment bias (W. H. Cooper, 1981) and less reliance on non-diagnostic cues (Dignath & Sprenger, 2020).

The present study aimed to assess the accuracy of teachers' SK judgments across various SRL components (cognitive strategies, metacognitive strategies, time management strategies, management of learning environment strategies, and motivation regulation strategies) by comparing teachers' judgments with reliable knowledge tests (Brunner et al., 2025). Based on our previous research on teachers' accuracy in SK judgments (Karlen, Bäuerlein et al., 2023) and results regarding their judgment accuracy of learning skills in general (Urhahne & Wijnia, 2021), we hypothesized that teachers would have low accuracy across all five SRL components (H1). The second aim was to examine potential biases in teachers' SK judgments with respect to students' academic achievement, gender, and SES. Based on previous research (Carr & Kurtz, 1991; Sneyers et al., 2020), we expected teachers to overestimate students with higher school grades (H2.1), female students (H2.2), and students from higher SES backgrounds (H2.3). Third, we examined various teacher characteristics as potential sources of interindividual differences in SK judgment biases across grades (H3.1), gender (H3.2), and SES (H3.3). We expected higher content and assessment knowledge of SRL to be associated with lower biases and, exploratorily, investigated the effects of teachers' diagnostic self-concept and diagnostic experiences. Lastly, we expected that stronger stereotypes favoring girls would be associated with greater overestimation of girls. Due to the lack of prior research, we did not formulate component-specific hypotheses. However, we examined all hypotheses separately for each SRL component.

## Methods

### Sample

Data collection took place in February and March 2022 as part of a longitudinal research project involving 1,295 students in lower secondary education. In the present study, we analyzed a subsample of students for whom their teachers judged their SK skills at the first measurement point. To minimize participant burden, teachers were asked to rate the SK skills of only four students per class, resulting in a potential sample of  $N=276$ . However, some teachers did not complete ratings for all four selected students, or the corresponding student SK scores were missing (e.g., because a student did not participate in the study or was absent during data collection). As a result, the number of valid matches between teacher judgments and student scores was further reduced ( $N=200$ ).

Thus, the current sample comprised 200 students and 69 teachers from 16 schools across two countries. Of these, 141 students and 47 teachers were from Switzerland, while 59 students and 22 teachers were from Germany. Secondary education in both countries is organized into three tracks of varying difficulty (basic, intermediate, and high), with placement determined by students' performance in primary school. Some schools also include classes with students from different track levels or differentiate levels by subject. The current sample included students from all three track levels (basic = 15.000%, intermediate = 33.500%, high = 23.000%, mixed or unknown = 28.500%). The students' mean age was 14.070 years (Range: 11.583–17.083), with a roughly even gender distribution (51.256% female, 48.241% male, 0.503% diverse). Students were primarily in 7th grade ( $n=115$ ) or 8th grade ( $n=66$ ), with a small number from 9th grade ( $n=19$ ). Teachers were predominantly female (65.217% female, 34.783% male) with a mean age of 39.892 years (Range: 24.583–64.333). Teachers had a mean work experience of 12.529 years, ranging from 0 to 38 years.

### Design

Teachers were instructed to complete the teacher questionnaire as an initial step. They could complete the questionnaire at any time during the weeks preceding students' SK tests. Students completed the SK tests and an online questionnaire assessing additional variables (e.g., demographics) during class under their teachers' supervision. The SK tests were developed as part of the larger research project for use in a digital SRL tool (Brunner et al., 2025; Karlen et al., 2026). As the larger research project focused on the effects of the digital tool, it included a control group. Therefore, some students in our sample completed the SK tests directly in the digital tool ( $n=169$ ), while others ( $n=31$ ) completed them in an online survey program. The primary difference between the two groups concerned the automated feedback regarding students' SRL, which the control group did not receive. Since teachers provided their judgments as part of the teacher questionnaire before students solved the SK tests, their judgments were unaffected by this difference. Therefore, the distinction between groups was deemed irrelevant for the present analysis.

The university's ethics committee approved the research project, from which the data for the current study were derived. Legal guardians of the participating students provided their consent, and students could withdraw from the study at any time without providing a reason or incurring any consequences.



## Instruments

### Teachers' SK judgments

To ensure informed judgments (Urhahne & Wijnia, 2021), teachers were first presented with example items from the SK tests to become familiar with their format and difficulty. Next, teachers were asked to judge the SK of the first four students on their class list by indicating the extent to which various statements applied to these students. Judgments were provided individually for each of the four students, with one statement per SRL component.

Table 1 provides an overview of the five SRL components and the corresponding items. The judgments were not tied to a specific situation or observation, as the focus was on SK, not on strategy use or SRL behavior.

### Teachers' content knowledge of SRL

We developed a test based on the measure by Clift et al. (1990) to assess teachers' content knowledge of strategies, a critical aspect of their content knowledge regarding SRL. Teachers read four scenarios describing different learning situations and had to choose the two most appropriate strategies for each scenario from a set of four to six strategies. They were instructed to select the strategies they would recommend to students in the situation described. For example, one scenario stated: "The class 6f is taking a biology test tomorrow. Today, the students have one more hour to prepare for it." Then, teachers had to choose two of the following strategies: (1) Generating questions and answering these. (2) Writing a summary. (3) Analyzing one's knowledge level. (4) Getting an overview. In this specific scenario, the correct solutions were 1 and 3. Although summarizing and getting an overview are generally effective strategies, they are more suitable for earlier stages of the learning process than for final test preparation. For each correctly selected or correctly rejected strategy (18 strategies overall), one point was awarded. The mean of all received points was calculated, yielding a score ranging from 0 to 1, with higher scores indicating greater knowledge.

**Table 1** SK judgments

| SRL component                          | Statement                                                                                        |
|----------------------------------------|--------------------------------------------------------------------------------------------------|
| SK of cognitive strategies             | He/She knows which learning strategies support information processing, storage, and retrieval.   |
| SK of metacognitive strategies         | He/She knows how to monitor and reflect on their learning purposefully.                          |
| SK of time management strategies       | He/She knows how to create an effective time and study plan.                                     |
| SK of learning environment strategies  | He/She knows how a workspace should be arranged to promote concentration and effective learning. |
| SK of motivation regulation strategies | He/She knows how to motivate themselves effectively for studying.                                |
| Rating Scale                           | 1 = does not apply, 2 = does rather not apply, 3 = does rather apply, 4 = does apply             |



## Teachers' assessment knowledge of SRL

This test was developed following the approach of Taxis et al. (2017), who analyzed traces in students' learning materials (e.g., side notes, highlights) as indicators of students' strategy use. We developed three tasks featuring fictional student learning material (including traces of SRL strategy use) and excerpts from the student's learning journal (before and after learning). Teachers were asked to diagnose the students' strategy use based on this information. An example of the developed learning traces is provided in Appendix A.

Teachers were asked to identify the strategies the student had successfully implemented and those that had been implemented poorly. They answered in an open-ended format and were asked to identify up to three well-executed and three poorly executed strategies, with three text boxes provided for each. Prior to administering the teachers' questionnaire, we defined, based on theory, which strategies were classified as well or poorly executed. According to the developed manual, teachers' responses were scored 0 or 1 for each named strategy using consensual coding. Strategies not listed in the established manual were reviewed by the research team and added if deemed appropriate. For each of the three tasks, a maximum of six points could be achieved. A sum score was calculated, resulting in a scale ranging from 0 to 18.

## Teachers' diagnostic experience regarding SRL

A single item measured teachers' previous experience with diagnosing SRL: "How much experience do you have in diagnosing self-regulated learning in class?" Responses were given on a four-point scale (1 = "(almost) no experience", 2 = "little experience", 3 = "some experience", 4 = "much experience").

## Teachers' diagnostic self-concept regarding SRL

Three items adapted from Retelsdorf et al. (2014) measured teachers' self-concept of their diagnostic competence (see Appendix A for details). An example item is: "I am good at assessing students' competencies in self-regulated learning." Items were rated on a six-point scale (1 = "Does not apply at all" to 6 = "Applies completely"). A mean score was calculated, resulting in a score ranging from 1 to 6. The scale's internal consistency was good ( $\alpha=0.828$ ,  $\omega=0.833$ ).

## Teachers' gender stereotypes in SRL

Teachers' gender stereotypes were assessed using a self-developed scale based on that of Retelsdorf et al. (2015). It included three items (provided in Appendix A) that assessed teachers' gender stereotypes regarding SRL. One example is: "Who can work more independently on long-term projects?" Items were rated on a scale from 1 to 5 (1 = "boys much better", 2 = "boys better", 3 = "both the same", 4 = "girls better", 5 = "girls much better"). The internal consistency was at the lower threshold of acceptability ( $\alpha=0.693$ ,  $\omega=0.714$ ).

## Students' SK tests

All five SK tests were developed within the larger research project and initially implemented in a pilot study with a separate student sample (Karlen, Bäuerlein et al., 2023). In the next step, the tests were revised and then tested and validated in another study (Brunner et al., 2025), which demonstrated acceptable reliability, a unidimensional factor structure, and theoretically consistent associations between the SK tests and student characteristics. A subsequent study provided further information regarding the test development process (Karlen et al., 2026). These samples partially overlapped with the current sample.

Internal consistency within the sample of the current study was assessed using Cronbach's alpha (Cronbach, 1951) and McDonald's omega (McDonald, 1999) and reported in the following detailed descriptions of each test. Alpha values above 0.70 and omega values above 0.65 are generally considered indicative of acceptable reliability (Kalkbrenner, 2023). However, the inclusion of various task formats and the assessment of heterogeneous constructs can reduce internal consistency in knowledge tests without necessarily indicating low reliability (Edelsbrunner et al. 2025a, b). Thus, using strict cutoff values is not recommended (Edelsbrunner et al. 2025b). Nevertheless, all SK tests met our expectations for omega values, although some of the alpha values were below the conventional thresholds. The unidimensional factor structure of the SK tests was further evaluated using confirmatory factor analyses in the R package lavaan (Rosseel, 2012) with robust maximum likelihood estimation. SRMR < 0.08, RMSEA < 0.06, and TLI/CFI > 0.95 indicate adequate model fit (Hu & Bentler, 1999). Given the relatively small sample size for CFAs, fit indices should be interpreted with caution. In particular, TLI and RMSEA are known to reject acceptable models too often in smaller samples (Hu & Bentler, 1999). Chi-square tests indicated no significant differences between the unidimensional models and the underlying data, except for the SK test of motivation regulation. Furthermore, the correlation matrix (Table 3, presented in the results section) showed positive associations between the SK test values and students' achievement, SES, and female gender. These associations correspond to those expected from previous studies (Händel et al., 2013; Karlen et al., 2014), thereby supporting the validity of the SK tests.

Consistent with teachers' SK judgments, the SK tests assessed domain-general yet component-specific SK. While most tasks described specific learning situations, the subjects and contexts varied within a test. Such scenario-based tests have been used in research on students' strategy knowledge (Dörrenbächer-Ulrich et al., 2024; Händel et al., 2014) to assess students' knowledge of effective strategies across different learning situations (Flavell, 1979). The assessed SK does not necessarily translate into students' actual use of these strategies during learning. Instead, it represents their underlying knowledge about these.

The SK tests consisted of 5–9 tasks, some of which comprised multiple subtasks. It is important to note that all tasks were knowledge-based and did not rely on self-assessment. The formats of the tasks differed widely (e.g., selecting the best alternative, rating the usefulness of various strategies, sorting tasks). For each task, students could receive 0–1 points. Most tasks were scored according to theoretically defined correct responses. However, we relied on expert ratings by researchers in the field of SRL ( $N = 14$ – $15$ ) for tasks that included pair comparisons. These tasks primarily involved rating the usefulness of various SRL strategies in a fictional learning situation. Students' ratings of the relative usefulness of strategies (e.g., strategy A as more useful than strategy B) were compared with those of experts,

with 1 point awarded when the ratings aligned. To compute the task score (mean value) for each scenario, we included only pair comparisons in which 75% of the experts agreed that one strategy was superior to the other. To calculate the SK test scores for each SRL component, the means of the corresponding task scores were computed (0–1) and converted to test scores ranging from 0 to 100, with higher scores indicating higher SK. For each SRL component, an example task is provided in Appendix A.

The *SK test of cognitive strategies* consisted of 9 tasks that primarily focused on elaboration, organization, and repetition strategies. Some tasks required students to evaluate whether a strategy was implemented appropriately. For example, students were required to identify key aspects of effective summaries. This knowledge is relevant because even undergraduate students often struggle to write effective summaries, thereby limiting the potential benefits of this strategy (Spiegel & Delaney, 2016). In other tasks, students had to select appropriate cognitive strategies for specific learning situations. For instance, students had to rate the usefulness of various reading strategies to support retention, indicating their understanding of the complex interplay between reading strategies and situational contexts (Flippo & Caverly, 2009). In another task, students had to choose the two most effective strategies from four to learn words in a second language, in a way that would facilitate their use in role-play. Thus, tasks included different formats and assessed SK of various cognitive strategies, resulting in internal consistency that was barely acceptable ( $\alpha=0.695$ ,  $\omega=0.739$ ). The unidimensional model showed good fit (CFI=1.000, TLI=1.027, RMSEA=0.000, SRMR=0.040) and an insignificant chi-square value ( $p=0.705$ ).

The *SK test of metacognitive strategies* consisted of five learning scenarios focusing on monitoring and reflection strategies (e.g., a fictional student writing an essay). Students rated the usefulness of up to six presented strategies for each scenario on a four-point scale. Some options reflected the absence of strategy use (e.g., the student does not revise the essay), while other options included unstructured monitoring or reflection (e.g., the student revises the essay without a plan) or structured monitoring (e.g., the student revises the essay purposefully). Students' ratings of the strategies' usefulness were compared to expert ratings, following the procedure outlined earlier. Internal consistency was acceptable to good ( $\alpha=0.779$ ,  $\omega=0.817$ ). The unidimensional model fit was adequate (CFI=0.985, TLI=0.970, RMSEA=0.060, SRMR=0.030) and the chi-square test was insignificant ( $p=0.133$ ).

The *SK test of time management strategies* consisted of nine tasks assessing knowledge of effective time management. In some tasks, students had to choose the most useful time plan or select the most helpful time-planning tip, reflecting knowledge of buffer times and the distribution of study time (Latimier et al., 2021). In other tasks, various to-dos had to be sorted in the order students recommended completing them, which required knowledge of prioritization based on urgency and importance (Murray et al., 2022). Additionally, the test assessed students' knowledge of selecting appropriate time-planning goals, a crucial aspect of time management (R. V. Adams & Blair, 2019). In these tasks, the most useful goal had to be chosen out of four options (e.g., "I will do my best while learning", "I will solve five tasks today and on Wednesday. On Friday, I will repeat everything again"). The internal consistency of this SK test was relatively low ( $\alpha=0.589$ ,  $\omega=0.650$ ), which may be attributable to the wide range of time management aspects and task formats included (Edelsbrunner et al. 2025a). This corresponds to the partially insufficient fit values of the unidimensional model (CFI=0.912, TLI=0.882, RMSEA=0.044, SRMR=0.056). However, the

chi-square test indicated no significant difference between the model and the underlying data ( $p=0.087$ ), indicating a reasonable fit.

The *SK test of learning environment strategies* consisted of eight tasks assessing knowledge of structuring the environment to support learning (Rost, 2018). This encompassed knowledge of factors such as fresh air (Wargocki et al., 2020) and the minimization of distractions (Duckworth et al., 2014). In one task, four graphical illustrations of a desk were presented, and students were required to select the desk that would be best for learning. In other tasks, students were asked to select the best of three options to structure the learning environment for fictional students (e.g., opening the window from time to time for fresh air, keeping the window closed to avoid disrupting study time, or leaving the window open all the time despite the disturbing noise from outside). Internal consistency was low ( $\alpha=0.571$ ,  $\omega=0.678$ ) along with certain unsatisfactory fit indices of the unidimensional model (CFI=0.907, TLI=0.869, RMSEA=0.047, SRMR=0.055). However, the chi-square test indicated no significant differences between the model and the underlying data ( $p=0.111$ ), suggesting an acceptable fit.

The *SK test of motivation regulation strategies* consisted of six tasks in which students rated the usefulness of various motivation regulation strategies for a specific scenario. The strategy options ranged from no systematic strategy use (e.g., postpone studying until the next day), to mismatched strategies that did not address the underlying motivational problem (e.g., enhancing the learning environment when the underlying issue was fear and overwhelm), to targeted strategies that directly addressed the motivational deficit (e.g., using positive self-talk or seeking social support to manage fear). Thus, the tests assessed students' knowledge about the fit of a strategy to a specific situation (Miele et al., 2024). Pairwise comparisons between strategies were used to assign points based on students' ratings relative to expert ratings. Internal consistency was high ( $\alpha=0.849$ ,  $\omega=0.892$ ). Still, fit indices of the unidimensional model showed a heterogeneous picture, with some acceptable and some unsatisfactory values (CFI=0.962, TLI=0.937, RMSEA=0.096, SRMR=0.046), and a significant chi-square result ( $p=0.007$ ).

## Students' school grades

Schools supplied students' report cards from the fall semester of 2021. Thus, these grades reflected students' academic achievement over the preceding half-year. In addition to the report cards, students self-reported their fall semester grades. Self-reported data were used when schools failed to provide report cards, which was the case for many schools in Germany. A composite score was calculated from students' grades in German, Math, and social/natural sciences, comparable to a grade point average (GPA), with higher grades reflecting higher academic achievement. As students receive grades from different subject teachers in secondary education, component scores likely reflect assessments from multiple teachers beyond those involved in the study. However, biannual grading conferences are standard in both countries, ensuring that all teachers are informed of students' overall achievement. Furthermore, the teachers involved in the study were generally class teachers, a group typically expected to have a comprehensive overview of their students' academic achievement to identify and support struggling students.

## Students' gender

Students indicated their gender in the additional online questionnaire as “female”, “male”, or “diverse”.

## Students' SES

Students self-reported the number of books in their homes as an indicator of SES in the additional online questionnaire. Answer options ranged from 1 (“none or very few”) to 6 (“enough to fill a massive shelf [more than 500 books]”) (PISA study; Mang et al., 2019), accompanied by illustrative images of bookshelves. Despite technological advances (e.g., e-books), this single item remains a reliable indicator of SES (Heppt et al., 2022).

## Analyses

All analyses were performed in the statistical program R (v.4.4.1; R Core Team, 2024). An analysis of missing data indicated a small percentage of missing values overall (2.8%), with the highest proportion in school grades (16%). This was due to most schools from Germany not providing the report cards. Consequently, we relied on self-reports for these students, which included some missing data. Similarly, some schools in Germany did not consistently complete the students' questionnaires, resulting in a higher proportion of missing values for the SES variable (12%).

ANOVA results (reported in Appendix B) revealed significant clustering of teachers' SK judgments (the dependent variables) for cognitive strategies, time management, and the learning environment by class, respectively, by teacher. Some SK test scores were also significantly clustered within classes. Thus, we controlled for class membership in all subsequent analyses.

Teachers' judgment accuracy can be operationalized as the difference between teachers' scores and students' test scores (absolute accuracy) or their correlation (relative accuracy), the latter indicating teachers' ability to rank students correctly (Urhahne & Wijnia, 2021). We focused on relative accuracy, as it enabled us to examine SK test scores and student characteristics simultaneously as predictors of teachers' SK judgments. This would not have been possible with a difference score, in which the test score is already embedded in the measure.

For H1, we assessed relative accuracy using correlation analyses with cluster-corrected standard errors in the lavaan package (Rosseel, 2012). The interpretation of observed correlation coefficients is not straightforward and depends on context, with varying threshold recommendations (Lovakov & Agadullina, 2021). Thus, we did not use strict thresholds but examined the descriptive correlations between teachers' SK judgments of the five SRL components and students' SK test scores. However, as an aid to interpretation, we relied on the commonly used thresholds of  $r=0.10$  for a small effect,  $r=0.30$  for a medium effect, and  $r=0.50$  for a large effect (Cohen, 1988).

We further investigated H1 and the remaining hypotheses (H2-3) in multilevel models using the lme4 package (Bates et al., 2015). First, an intercept-only model was estimated separately for each SRL component with teachers' SK judgments as the dependent variable (model 0). Next, models were estimated in which students' SK scores were used as the

predictor of their teachers' SK judgments (model 1). After this, students' grades, gender, and SES were added as predictors of teachers' SK judgments (model 2). For these analyses, gender was coded as a dichotomous variable (0=female, 1=male). Responses indicating a diverse gender identity were coded as missing due to the small number of cases, which precluded reliable estimation in the regression model. This approach ensured that these students remained in the overall analyses without influencing gender-related effects. If grades, gender, or SES predicted teachers' SK judgments beyond students' SK scores, this would indicate an overreliance on these characteristics, beyond their actual association with SK, and thus a biased judgment (H2). Semi-partial  $R^2$  (Jaeger, 2025; Nakagawa & Schielzeth, 2013) was computed as an indicator of the unique variance explained by each predictor. Since the predictors were also correlated with each other, generalized Variance Inflation Factors were calculated (Fox & Monette, 1992; Lüdtke et al., 2021). All values were below 2, indicating no problematic multicollinearity (Zuur et al., 2010). For each student characteristic, random slopes were tested separately using  $\chi^2$ -tests for model fit ( $-2 \log$  Likelihood values). The normality of residuals and random effects, as well as homoscedasticity, were tested for each reported model. Since assumptions were not fully met for some models, robust estimators were computed for the final model (model 2), using the *robustlmm* package (Koller, 2016) as an additional step.

Teachers' content and assessment knowledge of SRL, their diagnostic self-concept and diagnostic experience, and their gender stereotypes were tested as possible moderators of their judgment biases by adding cross-level interactions. Thus, we examined whether the effect of students' characteristics on teachers' judgments, while controlling for students' SK test scores, depended on teacher characteristics (H3). Interactions were further examined using simple slope analyses in the interaction package (Long, 2024), with adjustments for multiple comparisons (Esarey & Sumner, 2017).

For all models, student-level predictors were z-transformed (except for dichotomous variables) as a method of grand mean centering while standardizing at the same time. Final models and significant interactions were additionally estimated with predictors centered within clusters (CWC). Teacher-level predictors were grand-mean centered (centering in multilevel models; Enders & Tofghi, 2007).

## Results

### Descriptive statistics

Table 2 presents the distribution of the study variables. Students' SK test scores and teachers' SK judgments covered the complete scale range for most components, indicating substantial variability. The highest mean values were observed in the SK test of the learning environment, consistent with the pattern in teachers' SK judgments. Teachers expressed relatively high confidence in their diagnostic competence regarding SRL ( $M=4.140$  on a scale from 1 to 6), in contrast to their limited self-reported diagnostic experience ( $M=2.036$  on a scale from 1 to 4). Teachers' content and assessment knowledge of SRL showed considerable heterogeneity, with mean values above the scale means. The stereotype measure ranged from 3.000 (belief in equal SRL abilities between males and females) to 4.667 (belief in females' superiority in SRL). No teacher endorsed a stereotype favoring boys.

**Table 2** Distribution of study variables

| Variable | Min    | Q1     | Mdn    | M      | Q3     | Max     |
|----------|--------|--------|--------|--------|--------|---------|
| Student  |        |        |        |        |        |         |
| SK cog   | 20.833 | 44.907 | 56.481 | 57.829 | 72.421 | 93.519  |
| SK met   | 0.000  | 44.861 | 59.861 | 57.622 | 74.514 | 97.500  |
| SK time  | 12.593 | 51.667 | 63.333 | 63.611 | 77.037 | 100.000 |
| SK env   | 0.000  | 62.500 | 75.000 | 73.782 | 87.500 | 100.000 |
| SK mot   | 2.083  | 53.618 | 72.269 | 66.118 | 83.704 | 100.000 |
| Grades   | 1.000  | 4.333  | 4.625  | 4.572  | 5.000  | 6.000   |
| SES      | 1.000  | 2.000  | 3.000  | 3.120  | 4.000  | 6.000   |
| Teacher  |        |        |        |        |        |         |
| TJ cog   | 1.000  | 2.000  | 3.000  | 2.665  | 3.000  | 4.000   |
| TJ met   | 1.000  | 2.000  | 3.000  | 2.688  | 3.000  | 4.000   |
| TJ time  | 1.000  | 2.000  | 3.000  | 2.890  | 3.625  | 4.000   |
| TJ env   | 1.000  | 3.000  | 3.000  | 3.215  | 4.000  | 4.000   |
| TJ mot   | 1.000  | 2.000  | 3.000  | 2.735  | 3.000  | 4.000   |
| CK-SRL   | 0.222  | 0.667  | 0.778  | 0.745  | 0.833  | 1.000   |
| AK-SRL   | 2.000  | 7.000  | 11.000 | 10.203 | 12.250 | 17.000  |
| Dia SC   | 2.333  | 3.667  | 4.000  | 4.140  | 4.667  | 5.667   |
| Dia Exp  | 1.000  | 2.000  | 2.000  | 2.036  | 2.500  | 4.000   |
| Gen Ste  | 3.000  | 3.000  | 3.667  | 3.578  | 4.000  | 4.667   |

SK test scores: 0–100; Teacher judgments: 1–4. *SKcog* SK of cognitive strategies, *SKmet*SK of metacognitive strategies, *SKtime*SK of time management strategies, *SKenv*SK of learning environment strategies, *SKmot*SK of motivation regulation strategies, *SES*socio-economic status, *TJcog*teacher judgment of cognitive strategies, *TJ met*teacher judgment of metacognitive strategies, *TJtime*teacher judgment of time management, *TJenv*teacher judgment of learning environment, *TJmot*teacher judgment of motivation regulation, *CK-SRL* teacher content knowledge of SRL, *AK-SRL* teacher assessment knowledge of SRL *DiaSC* diagnostic self-concept, *Dia Exp* diagnostic experience, *Gen Ste*Gender stereotypes

## H1 - Teacher judgment accuracy

To examine teachers' relative judgment accuracy, we analyzed correlations between teachers' SK judgments and students' corresponding SK test scores, as reported in Table 3. These were low to moderate but significant across all SRL components ( $r=0.243$ – $0.344$ ). However, at a descriptive level, correlations between teachers' judgments and students' corresponding SK test scores were not systematically higher than cross-component correlations. This lack of component specificity is further supported by the relatively high intercorrelations between teachers' judgments ( $0.387$ – $0.746$ ), which exceeded those of students' SK test scores ( $0.363$ – $0.643$ ,  $M_{\text{diff}} = 0.104$ ).

Teachers' SK judgments were positively correlated with students' grades and exhibited a weak positive correlation with students' SES. Correlations with gender were negative, indicating higher SK judgments for female students. The same pattern was observed for students' SK test scores, which showed similar associations with grades, gender, and SES. However, teachers' SK judgments correlated more strongly with students' school grades than did their SK test scores ( $M_{\text{diff}} = 0.234$ ). Overall, the pattern of the correlation matrix suggests low accuracy in teachers' SK judgments, supporting H1.



**Table 3** Correlation matrix of study variables

|         | SK met    | SK time   | SK env   | SK mot   | Grades   | SES     | Gender    | TJ cog   | TJ met   |
|---------|-----------|-----------|----------|----------|----------|---------|-----------|----------|----------|
| SK cog  | 0.539***  | 0.476***  | 0.393*** | 0.540*** | 0.396*** | 0.297** | -0.293*** | 0.344*** | 0.366*** |
| SK met  |           | 0.489***  | 0.363*** | 0.643*** | 0.309**  | 0.132   | -0.157*   | 0.233**  | 0.243**  |
| SK time |           |           | 0.439*** | 0.495*** | 0.338*** | 0.183*  | -0.220**  | 0.256*** | 0.321*** |
| SK env  |           |           |          | 0.462*** | 0.356*   | 0.165*  | -0.215*** | 0.226**  | 0.324*** |
| SK mot  |           |           |          |          | 0.287**  | 0.161*  | -0.180**  | 0.135*   | 0.200*   |
| Grades  |           |           |          |          |          | 0.298** | -0.189**  | 0.604*** | 0.622*** |
| SES     |           |           |          |          |          |         | 0.027     | 0.110    | 0.217*   |
| Gender  |           |           |          |          |          |         |           | -0.081   | -0.198** |
| TJ cog  |           |           |          |          |          |         |           |          | 0.674*** |
| TJ met  |           |           |          |          |          |         |           |          |          |
| TJ time |           |           |          |          |          |         |           |          |          |
| TJ env  |           |           |          |          |          |         |           |          |          |
| TJ mot  |           |           |          |          |          |         |           |          |          |
| CK-SRL  |           |           |          |          |          |         |           |          |          |
| AK-SRL  |           |           |          |          |          |         |           |          |          |
| Dia SC  |           |           |          |          |          |         |           |          |          |
| Dia Exp |           |           |          |          |          |         |           |          |          |
|         | TJ env    | TJ mot    | CK-SRL   | AK-SRL   | Dia SC   | Dia Exp | Gen Ste   |          |          |
| SK cog  | 0.257***  | 0.328***  | 0.232*   | 0.003    | 0.075    | 0.158*  | -0.069    |          |          |
| SK met  | 0.135     | 0.238***  | 0.138    | -0.008   | 0.120    | 0.073   | -0.056    |          |          |
| SK time | 0.314***  | 0.276***  | 0.073    | 0.064    | 0.104    | 0.144   | 0.086     |          |          |
| SK env  | 0.288**   | 0.287***  | 0.048    | -0.033   | 0.052    | 0.001   | 0.072     |          |          |
| SK mot  | 0.206**   | 0.245***  | 0.091    | 0.091    | 0.075    | 0.103   | 0.003     |          |          |
| Grades  | 0.459***  | 0.587***  | 0.169    | 0.045    | 0.198    | 0.181   | -0.010    |          |          |
| SES     | 0.042     | 0.197*    | -0.082   | -0.097   | 0.021    | 0.072   | -0.134    |          |          |
| Gender  | -0.351*** | -0.257*** | -0.031   | -0.036   | -0.065   | 0.018   | -0.013    |          |          |
| TJ cog  | 0.387***  | 0.548***  | 0.150    | 0.008    | 0.145    | 0.203*  | -0.025    |          |          |
| TJ met  | 0.571***  | 0.587***  | 0.096    | -0.130   | 0.114    | 0.128   | 0.112     |          |          |

Table 3 (continued)

|         | TJ env   | TJ mot   | CK-SRL | AK-SRL | Dia SC | Dia Exp  | Gen Ste |
|---------|----------|----------|--------|--------|--------|----------|---------|
| TJ time | 0.674*** | 0.556*** | 0.127  | -0.059 | 0.161  | 0.117    | 0.085   |
| TJ env  |          | 0.557*** | 0.144  | 0.121  | 0.166  | 0.087    | 0.084   |
| TJ mot  |          |          | 0.071  | 0.044  | 0.154* | 0.123    | 0.096   |
| CK-SRL  |          |          |        | 0.260* | 0.123  | 0.198    | -0.067  |
| AK-SRL  |          |          |        |        | 0.220  | 0.163    | 0.048   |
| Dia SC  |          |          |        |        |        | 0.490*** | 0.042   |
| Dia Exp |          |          |        |        |        |          | 0.068   |

*SK cog* SK of cognitive strategies, *SK met* SK of metacognitive strategies, *SK time* SK of time management strategies, *SK env* SK of learning environment strategies, *SK mot* SK of motivation regulation strategies, *SE* Socio-economic status, *TJ cog* teacher judgment of cognitive strategies, *TJ met* teacher judgment of metacognitive strategies, *TJ time* teacher judgment of time management, *TJ env* teacher judgment of learning environment, *TJ mot* teacher judgment of motivation regulation, *CK-SRL* teacher content knowledge of SRL, *AK-SRL* teacher assessment knowledge of SRL, *Dia SC* diagnostic self-concept, *Dia Exp* diagnostic experience, *Gen Ste* Gender stereotypes. Gender was coded female=0; male=1. Cluster-corrected SEs were used. Teachers' self-concept regarding SRL diagnosis, their gender stereotype, and teachers' CK-SRL and AK-SRL were assessed per teacher, and thus, had identical values for all students from the same class. For correlations among only those variables, a dataset containing only the 69 teachers (compared to the 200 students) was used. For correlations between those variables and students' variables, the student dataset was used, where teachers' values were repeated for each of their students

\*  $p < 0.05$ . \*\*  $p < 0.01$ . \*\*\*  $p < 0.001$

## H2 - Teacher judgment predicted by student characteristics

Table 4 presents the results of the multilevel models predicting teachers' SK judgments. Model 0 represents the baseline model without predictors, followed by Model 1, in which the SK judgments were predicted solely from students' SK test scores. In the next step, student characteristics were added as predictors of teachers' SK judgments in Model 2.

In line with the previously observed correlations, students' SK test scores significantly predicted teachers' SK judgments in the multilevel models (Table 4, Model 1). Nevertheless, students' SK test scores explained less than 12% of the variance in teachers' SK judgments across components, as indicated by the marginal  $R^2$  values for Model 1. Additionally, when student characteristics were included as predictors (Table 4, Model 2), the effects of the SK test scores became nonsignificant. This pattern was observed across all components. Furthermore, teachers' SK judgments were mainly predicted by students' grades rather than students' SK test scores, indicating a substantial achievement bias and supporting H2.1. Gender (H2.2) significantly predicted teachers' judgments of SK of the learning environment and SK of motivation regulation, while controlling for students' SK test scores, school grades, and SES. Thus, teachers rated a female student's SK higher than a male student's SK, even if their performance in the SK tests was comparable. For teachers' judgments of students' SK of metacognitive strategies and time management, the gender effect was not significant, but a trend in the same direction was observed. No significant SES effect emerged for any SRL component, refuting H2.3.

Adding student characteristics as predictors substantially improved model fit statistics and explained variances. In those models, fixed effects accounted for 34.1% to 43.7% of the variance in teachers' SK judgments (marginal  $R^2$ ). Among the predictors included, school grades accounted for the largest unique proportion of explained variance (semi-partial  $R^2$ ) across all SRL components. The relatively small intercept variance and the non-significant random slopes for student characteristics indicate that most of the variability in SK judgments occurred within teachers rather than between them. As a result, there was only limited variance at the teacher level that could potentially be explained by teacher-level predictors. Nevertheless, ICC values indicate that some variance between teachers was present.

## H3 – Interindividual differences in teachers' judgment biases

### H3.1 – Differences in achievement biases

Teacher variables did not account for teachers' reliance on grades when judging students' SK in cognitive, metacognitive, learning environment management, or motivation regulation strategies. However, teachers with greater content knowledge of SRL tended to show a stronger achievement bias when judging students' SK of time management strategies ( $B=0.579$ ,  $SE=0.314$ ,  $t=1.844$ ,  $p=0.067$ ). However, including the interaction term in the model yielded only a marginal improvement in explained variance, increasing from 50.0% to 51.5%.

**Table 4** Teacher estimation predicted by SK test scores and student characteristics

| <b>SK of cognitive strategies</b>     |                        |                        |                             |                        |          |
|---------------------------------------|------------------------|------------------------|-----------------------------|------------------------|----------|
|                                       | Model 0                | Model 1                | Model 2                     | Model 2 robust         |          |
| <i>n</i> level 1                      | 200                    | 183                    | 156                         | 156                    |          |
| <i>n</i> level 2                      | 69                     | 68                     | 58                          | 58                     |          |
| <i>Fixed effects</i>                  | <i>B (SE)</i>          | <i>B (SE)</i>          | <i>B (SE)</i>               | <i>B (SE)</i>          | <i>t</i> |
| Intercept                             | 2.659 (0.069)          | 2.741 (0.063)          | 2.736 (0.076)               | 2.586, 2.887           | 34.599   |
| SK                                    |                        | 0.261 (0.053)***       | 0.088 (0.057)               | -0.024, 0.199          | 1.496    |
| Grades                                |                        |                        | 0.494 (0.059)***            | 0.377, 0.609           | 7.604    |
| Gender                                |                        |                        | -0.059 (0.052)              | -0.201, 0.200          | 0.113    |
| SES                                   |                        |                        | -0.001 (0.101)              | -0.162, 0.043          | -0.981   |
| <i>Random effects</i>                 | Variance ( <i>SD</i> ) | Variance ( <i>SD</i> ) | Variance ( <i>SD</i> )      | Variance ( <i>SD</i> ) |          |
| Intercept                             | 0.136 (0.368)          | 0.106 (0.325)          | 0.056 (0.237)               | <0.001 (<0.001)        |          |
| Residuals                             | 0.518 (0.720)          | 0.418 (0.647)          | 0.309 (0.556)               | 0.419 (0.648)          |          |
| Model fit                             |                        |                        |                             |                        |          |
| AIC                                   | 480.215                | 402.119                | 295.854                     |                        |          |
| BIC                                   | 490.110                | 414.957                | 317.203                     |                        |          |
| R <sup>2</sup> conditional            | 0.207                  | 0.293                  | 0.475                       |                        |          |
| R <sup>2</sup> marginal               | 0.000                  | 0.115                  | 0.379                       |                        |          |
| ICC adjusted                          | 0.207                  | 0.202                  | 0.153                       |                        |          |
| <b>SK of metacognitive strategies</b> |                        |                        |                             |                        |          |
|                                       | Model 0                | Model 1                | Model 2                     | Model 2 robust         |          |
| <i>n</i> level 1                      | 200                    | 196                    | 163                         | 163                    |          |
| <i>n</i> level 2                      | 69                     | 69                     | 59                          | 59                     |          |
| <i>Fixed effects</i>                  | <i>B (SE)</i>          | <i>B (SE)</i>          | <i>B (SE)</i>               | <i>B (SE)</i>          | <i>t</i> |
| Intercept                             | 2.688 (0.062)          | 2.699 (0.060)          | 2.801 (0.070)               | 2.803 (0.072)          | 39.018   |
| SK                                    |                        | 0.204 (0.056)***       | 0.050 (0.051)               | 0.027 (0.054)          | 0.495    |
| Grades                                |                        |                        | 0.536 (0.059)***            | 0.554 (0.063)          | 8.754    |
| Gender                                |                        |                        | -0.178 (0.095) <sup>a</sup> | -0.164 (0.102)         | -1.606   |
| SES                                   |                        |                        | 0.032 (0.050)               | 0.044 (0.053)          | 0.829    |

Table 4 (continued)

| <i>Random effects</i>                   |               | Variance (SD)    | Variance (SD)               | Variance (SD)          | Variance (SD)   |
|-----------------------------------------|---------------|------------------|-----------------------------|------------------------|-----------------|
| Intercept                               |               | 0.059 (0.243)    | 0.053 (0.230)               | 0.031 (0.176)          | <0.001 (<0.001) |
| Residuals                               |               | 0.580 (0.762)    | 0.534 (0.731)               | 0.321 (0.567)          | 0.387 (0.622)   |
| Model fit                               |               |                  |                             |                        |                 |
| AIC                                     |               | 482.309          | 458.217                     | 305.107                |                 |
| BIC                                     |               | 492.204          | 471.330                     | 326.764                |                 |
| R <sup>2</sup> conditional              |               | 0.092            | 0.150                       | 0.487                  |                 |
| R <sup>2</sup> marginal                 |               | 0.000            | 0.066                       | 0.437                  |                 |
| ICC adjusted                            |               | 0.092            | 0.090                       | 0.088                  |                 |
| <b>SK of time management strategies</b> |               |                  |                             |                        |                 |
| Model 0                                 |               | Model 1          |                             | Model 2                |                 |
| <i>n</i> level 1                        | 200           | 197              | 164                         |                        |                 |
| <i>n</i> level 2                        | 69            | 69               | 59                          |                        |                 |
| <i>Fixed effects</i>                    |               | <i>B (SE)</i>    |                             | <i>B (SE)</i>          |                 |
| Intercept                               | 2.885 (0.068) | 2.897 (0.064)    | 3.080 (0.081)               |                        |                 |
| SK                                      |               | 0.286 (0.058)*** | 0.084 (0.057)               |                        |                 |
| Grades                                  |               |                  | 0.445 (0.056)***            |                        |                 |
| Gender                                  |               |                  | −0.200 (0.106) <sup>b</sup> |                        |                 |
| SES                                     |               |                  | 0.011 (0.054)               |                        |                 |
| <i>Random effects</i>                   |               | Variance (SD)    |                             | Variance (SD)          |                 |
| Intercept                               | 0.094 (0.307) | 0.075 (0.274)    | 0.088 (0.297)               |                        |                 |
| Residual                                | 0.622 (0.789) | 0.560 (0.749)    | 0.353 (0.564)               |                        |                 |
| Model fit                               |               |                  |                             |                        |                 |
| AIC                                     | 503.199       | 475.012          | 338.831                     |                        |                 |
| BIC                                     | 513.194       | 488.145          | 360.530                     |                        |                 |
| R <sup>2</sup> conditional              |               |                  | 0.500                       |                        |                 |
| R <sup>2</sup> marginal                 | 0.132         | 0.219            | 0.375                       |                        |                 |
|                                         | 0.000         | 0.114            |                             |                        |                 |
|                                         |               |                  |                             | Model 2 robust         |                 |
|                                         |               |                  |                             | 164                    |                 |
|                                         |               |                  |                             | 59                     |                 |
|                                         |               |                  |                             | <i>B (SE)</i>          |                 |
|                                         |               |                  |                             | 3.094 (0.080)          |                 |
|                                         |               |                  |                             | 0.086 (0.062)          |                 |
|                                         |               |                  |                             | 0.444 (0.061)          |                 |
|                                         |               |                  |                             | −0.209 (0.116)         |                 |
|                                         |               |                  |                             | −0.007 (0.059)         |                 |
|                                         |               |                  |                             | Variance (SD)          |                 |
|                                         |               |                  |                             | <0.001 (<0.001)        |                 |
|                                         |               |                  |                             | 0.480 (0.693)          |                 |
|                                         |               |                  |                             | <i>spR<sup>2</sup></i> |                 |
|                                         |               |                  |                             | 0.013                  |                 |
|                                         |               |                  |                             | 0.271                  |                 |
|                                         |               |                  |                             | 0.021                  |                 |
|                                         |               |                  |                             | 0.000                  |                 |
|                                         |               |                  |                             | <i>t</i>               |                 |
|                                         |               |                  |                             | 38.505                 |                 |
|                                         |               |                  |                             | 1.396                  |                 |
|                                         |               |                  |                             | 7.247                  |                 |
|                                         |               |                  |                             | −1.804                 |                 |
|                                         |               |                  |                             | −0.127                 |                 |

Table 4 (continued)

|                            | ICC adjusted                                  | 0.132            | 0.118 | 0.199                  |  |  |                |                        |        |
|----------------------------|-----------------------------------------------|------------------|-------|------------------------|--|--|----------------|------------------------|--------|
|                            | <b>SK of learning environment strategies</b>  |                  |       |                        |  |  |                |                        |        |
|                            | Model 0                                       | Model 1          |       | Model 2                |  |  |                | Model 2 robust         |        |
| n level 1                  | 200                                           | 195              |       | 166                    |  |  |                | 166                    |        |
| n level 2                  | 69                                            | 67               |       | 59                     |  |  |                | 59                     |        |
| Fixed effects              | B ( <i>SE</i> )                               | B ( <i>SE</i> )  |       | B ( <i>SE</i> )        |  |  |                | B ( <i>SE</i> )        | t      |
| Intercept                  | 3.214 (0.065)                                 | 3.216 (0.064)    |       | 3.526 (0.081)          |  |  | 95% CI         | 3.531 (0.077)          | 45.728 |
| SK                         | Variance ( <i>SD</i> )                        | 0.235 (0.056)*** |       | 0.059 (0.054)          |  |  |                | spR <sup>2</sup>       | 1.265  |
| Grades                     |                                               |                  |       | 0.321 (0.056)***       |  |  | -0.048, 0.168  | 0.067 (0.053)          | 5.911  |
| Gender                     |                                               |                  |       | -0.541 (0.103)***      |  |  | 0.210, 0.432   | 0.322 (0.054)          | -4.815 |
| SES                        |                                               |                  |       | -0.027 (0.053)         |  |  | -0.743, -0.335 | -0.482 (0.100)         | -0.258 |
| Random effects             | Variance ( <i>SD</i> )                        |                  |       | Variance ( <i>SD</i> ) |  |  | -0.131, 0.079  | -0.013 (0.051)         |        |
| Intercept                  | 0.089 (0.298)                                 | 0.081 (0.284)    |       | 0.103 (0.321)          |  |  |                | Variance ( <i>SD</i> ) |        |
| Residual                   | 0.574 (0.758)                                 | 0.538 (0.734)    |       | 0.340 (0.583)          |  |  |                | 0.078 (0.279)          |        |
| Model fit                  |                                               |                  |       | 0.340 (0.583)          |  |  |                | 0.311 (0.557)          |        |
| AIC                        | 487.700                                       | 464.496          |       | 341.181                |  |  |                |                        |        |
| BIC                        | 497.595                                       | 477.588          |       | 362.965                |  |  |                |                        |        |
| R <sup>2</sup> conditional | 0.134                                         | 0.202            |       | 0.494                  |  |  |                |                        |        |
| R <sup>2</sup> marginal    | 0.000                                         | 0.082            |       | 0.341                  |  |  |                |                        |        |
| ICC adjusted               | 0.134                                         | 0.131            |       | 0.232                  |  |  |                |                        |        |
|                            | <b>SK of motivation regulation strategies</b> |                  |       |                        |  |  |                |                        |        |
|                            | Model 0                                       | Model 1          |       | Model 2                |  |  |                | Model 2 robust         |        |
| n level 1                  | 200                                           | 193              |       | 162                    |  |  |                | 162                    |        |
| n level 2                  | 69                                            | 69               |       | 59                     |  |  |                | 59                     |        |
| Fixed effects              | B ( <i>SE</i> )                               | B ( <i>SE</i> )  |       | B ( <i>SE</i> )        |  |  |                | B ( <i>SE</i> )        | t      |
| Intercept                  | 2.738 (0.062)                                 | 2.753 (0.058)    |       | 2.898 (0.073)          |  |  | 95% CI         | 2.881 (0.080)          | 36.243 |
| SK                         | Variance ( <i>SD</i> )                        | 0.197 (0.057)*** |       | 0.080 (0.055)          |  |  | -0.030, 0.189  | 0.086 (0.060)          | 1.424  |
|                            |                                               |                  |       |                        |  |  |                | spR <sup>2</sup>       | 0.013  |





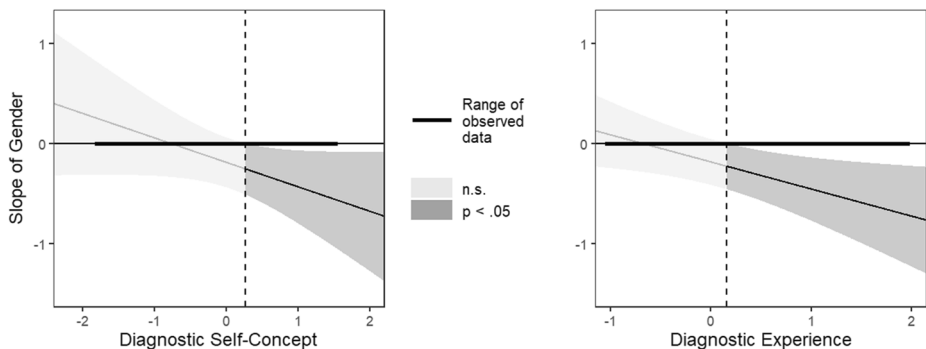
### H3.2 – Differences in gender biases

Teacher variables did not explain the gender bias in teachers' judgments of students' SK of the learning environment management strategies, motivation regulation strategies, or cognitive strategies.

However, for time management strategies, teachers who reported more diagnostic experience ( $B = -0.271$ ,  $SE = 0.105$ ,  $t = -2.583$ ,  $p = 0.011$ ) and a higher diagnostic self-concept ( $B = -0.244$ ,  $SE = 0.116$ ,  $t = -2.104$ ,  $p = 0.037$ ) exhibited a stronger bias favoring girls, although the gender bias across all teachers was only marginally significant. Detailed results of these models are reported in Appendix B. Simple slope analyses revealed a significant gender bias favoring females among teachers with values exceeding 0.156 for self-reported experience and 0.269 for diagnostic self-concept. Since both variables were grand-mean centered, this indicates that teachers with slightly above-average diagnostic experience and self-concept showed significant gender biases in their judgments of students' SK of time management (see Fig. 1). A similar trend of increased gender bias favoring females in teachers' judgments of SK of time management was evident among teachers with higher assessment knowledge of SRL ( $B = -0.038$ ,  $SE = 0.021$ ,  $t = -1.793$ ,  $p = 0.076$ ).

### H3.3 – Differences in SES Biases

No significant cross-level interactions between teacher variables and students' SES were found in predicting teachers' SK judgments.



**Fig. 1** Cross-level interactions for teachers' SK judgments of time management strategies

## Discussion

Students' SRL is connected to their teachers' SRL, which affects their promotion and, in turn, their students' SRL (Karlen et al., 2024). However, to enable adaptive promotion, accurate teacher judgments of students' SRL are critical. Thus, evaluating teachers' judgment accuracy is essential. In addition to examining teachers' judgment accuracy, the current study investigated the effects of students' characteristics on teachers' SK judgments and whether teachers' knowledge of SRL and their beliefs influence these judgments. The results revealed that teachers' SK judgments correlated weakly with students' SK test scores and were primarily predicted by students' school grades. This indicates that teachers rely on factors beyond diagnostic cues when evaluating students' SRL (Tannert et al., 2026). The following sections discuss the findings and their implications in more detail.

### Teacher judgment accuracy

Relative accuracy was assessed by examining the correlation between teachers' SK judgments and their students' SK test scores. The observed low-to-moderate correlations and limited component specificity in teachers' SK judgments indicated low judgment accuracy, thereby supporting H1. Despite this, teachers' accuracy remained higher than in our previous study, based on pilot project data (Karlen, B uerlein et al., 2023). The observed increase in accuracy could stem from revisions to the SK tests since the pilot study, which have enhanced test quality and potentially the consistency with teachers' judgments. Furthermore, in contrast to the pilot study, teachers in the current study were presented with example tasks from students' SK tests to ensure informed judgments, which typically yield higher accuracy than uninformed judgments (Urhahne & Wijnia, 2021). The observed correlations in the current study were comparable to those found for teachers' judgments of students' learning motivation or learning skills (Urhahne & Wijnia, 2021). However, teachers' judgment accuracy regarding SRL is clearly below the level typically found for academic achievement, estimated at  $r = 0.63$  in a meta-analysis based on 75 studies (S dkamp et al., 2012, p. 750). This meta-analytic estimate therefore provides a meaningful benchmark to contextualize the observed effects of the current study and underscores that judging students' SK may be especially difficult for teachers. Whether teachers would be more accurate in judging other SRL skills that might be more observable, such as strategy use, remains an open question. Furthermore, uncertainty persists about the required level of judgment accuracy to effectively adapt SRL promotion to students' needs. Therefore, the association between teachers' judgments and their promotion of SRL warrants further investigation.

### Biases in teachers' judgments

In the multilevel analyses, the initial association between students' SK test scores and teachers' SK judgments disappeared after adding student characteristics as predictors. This suggests that the commonly used correlation-based measure of relative accuracy (Urhahne & Wijnia, 2021) may overestimate teachers' competence in judging SK. Our results imply that the apparent relative accuracy may be primarily driven by teachers' reliance on students' school grades (H2.1) and gender (H2.2), which are linked to SK (Liu et al., 2021; Richardson et al., 2012). In contrast, students' SES did not impact teachers' SK judgments, contradicting H2.3. Thus, the multilevel models revealed that teachers tended to rate students with identical SK

test scores differently, depending on the students' grade and gender. However, what are the underlying reasons for these consistent biases in teachers' SK judgments? The substantially higher intercorrelations among teachers' SK judgments than among the SK test scores indicate a halo effect (Bechger et al., 2010; W. H. Cooper, 1981). Teachers may have judged students' SK based on their overall impression of the students rather than their specific SK, leading to inflated consistency across judgments. It is plausible that girls who tend to be more conscientious (Verbree et al., 2023) and students with higher grades were viewed more favorably by their teachers. Therefore, both student characteristics could have contributed to teachers' overall impressions of students. Additionally, these traits tend to be more salient than cues diagnostic of SK, like the quality of learning plans or students' cognitive processes. Conversely, students' SES might have been less salient to teachers. Reduced parent-teacher interactions in secondary education (K. S. Adams & Christenson, 2000; Kely & Wakabayashi, 2020) may restrict teachers' knowledge of students' family backgrounds. This could explain why the SES effect on teachers' judgments tends to be less pronounced among secondary school students than among younger students (Karing et al., 2024). Although SES did not affect teachers' SK judgments in our study, it could still have affected students' SK (Karlen et al., 2014).

Our findings align with previous research showing halo effects in educational settings on teachers' judgments of students' academic achievement (Dompnier et al., 2006; Schmidt et al., 2023), motivation (Urhahne et al., 2013), and engagement (Kaiser et al., 2013). We could extend this research to demonstrate indicators of a halo effect in SRL judgments as well. These converging results indicate that halo effects represent a broader pattern in teachers' judgments and are not limited to a single domain. Since teachers' judgments of students' current competence and their expectations for students' future development are closely intertwined (Zhu et al., 2018), this could potentially affect students' subsequent development, consistent with a self-fulfilling prophecy (H. M. Cooper, 1979). Specific to SRL, gender biases in teachers' judgments may perpetuate gender differences in SRL present throughout all age groups (Bidjerano, 2005; Händel et al., 2013; Matthews et al., 2009). Furthermore, achievement biases could reinforce SRL among high-achieving students and undermine it among low-achieving students, for example, by providing more learning opportunities to those students perceived as proficient in SRL.

## The effect of teacher characteristics on their judgment biases

### Knowledge, experience, and self-concept

The observed achievement and gender biases were mostly independent of teachers' characteristics. Cross-level interactions with teacher variables were mostly insignificant (H3.1–3). This is consistent with previous research that found no effects (Karlen, Bäuerlein et al., 2023; Tannert et al., 2026; Karing et al., 2024) or mixed results (Urhahne & Wijnia, 2021) regarding moderators of teachers' judgment accuracy.

Interestingly, the few interaction effects that emerged contradicted our theoretical expectations, indicating more pronounced biases among teachers with greater knowledge, a higher diagnostic self-concept, and more diagnostic experience. These effects were not consistently significant and were observed only for certain SRL components. Given the risk of inflated alpha error rates resulting from multiple interaction terms, these results should be interpreted cautiously. Nevertheless, the results suggest that greater knowledge does not

uniformly enhance recognition of diagnostic cues (Tannert et al., 2026), diverging from prior findings linking knowledge to improved awareness of diagnostic cues (Dignath & Sprenger, 2020) and higher diagnostic competence (Klug et al., 2016). This contradiction could be explained by two distinct processes underlying decision-making, either a reflective, analytical process or an intuitive one (Evans, 2010). Teachers with greater experience or knowledge of SRL may perform better on tasks that elicit reflective reasoning or structured assessments (Dignath & Sprenger, 2020; Klug et al., 2016; Kosel et al., 2024), whereas they may be more prone to relying on intuition in everyday classroom contexts. This aligns with previous results that a stronger halo bias in teachers' achievement judgments was associated with higher judgment certainty (Sanrey et al., 2021).

## Stereotypes

Contrary to our expectations, a stronger stereotype of female superiority in SRL was not associated with greater overestimation of females' SK. The stereotype measure may reflect teachers' accurate understanding that girls tend to perform better in SRL (Bidjerano, 2005; Liu et al., 2021), as was also evident in the current study. In this case, their "stereotype" would reflect reality and thus not result in overestimating females' SK when controlling for SK test scores. In other words, the measure of stereotypes might have unintentionally assessed teachers' knowledge about the association between gender and SRL. Alternatively, teachers with gender stereotypes may have used stereotype-based standards in their judgments of students' SRL. According to the "shifting standard theory," people tend to apply different standards depending on the group being judged (Biernat et al., 1991). Thus, teachers would unconsciously rate boys' performance as "good" relative to other boys, and girls' performance as "good" relative to other girls, masking the impact of stereotypes on their judgments.

## Practical implications

More than 30 years ago, Carr and Kurtz (1991) demonstrated a systematic achievement bias in teachers' judgments of students' SK, attributions, and self-concept. Despite the increased focus on SRL in education (OECD, 2019) as a prerequisite of lifelong learning (Van den Broeck et al., 2024), teachers still rely heavily on students' school grades and, to a lesser extent, on gender when diagnosing SK. These persistent biases, independent of teachers' SRL knowledge, imply that knowledge alone is insufficient to ensure diagnostic accuracy. This finding is consistent with experimental results showing that targeted information designed to enhance participants' knowledge of the Halo effect failed to prevent it (Wetzel et al., 1981). However, informing teachers of the presence of non-diagnostic cues increased their SRL judgment accuracy in an experimental setting (Tannert et al., 2026). Moreover, demonstrating the Halo effect to pre-service teachers and thereby exposing their own biased judgments increased their motivation to engage in systematic thinking and methodological research (Behrmann, 2019). This approach may promote teachers' awareness and motivate them to conduct systematic SRL assessments using tools such as rubrics. Rubrics support the assessment of complex skills or knowledge by specifying relevant indicators and how to judge their quality, by describing different performance or knowledge levels (Jönsson & Panadero, 2017). They have been shown to reduce biased judgments (Quinn, 2020). However, rubrics, questionnaires, or checklists to support reflective SRL assessment are seldom

used by teachers (Bäuerlein et al., 2023). Digital assessment tools could further support and complement teachers' SK judgments (van Leeuwen et al., 2022).

Given the complexity of SRL assessments and teachers' lack of relevant knowledge (Tannert et al., 2026), brief teacher training is unlikely to yield measurable improvement (Bäuerlein et al., 2023). Moreover, teachers' overconfidence in their diagnostic competencies may diminish motivation to pursue further teacher training. This highlights the importance of systematically integrating SRL and its assessment into teacher education programs. In general, the focus should extend beyond the provision of knowledge to include concrete assessment methods and an examination of the diagnostic validity of various cues (Tannert et al., 2026).

## Limitations and directions for future research

A challenge in accuracy research is determining the reference criterion for comparison with teachers' judgments (Karing et al., 2024), which is then considered the "correct" score. Especially when using self-reports, it may be more appropriate to refer to "agreement" between teacher and student rather than accuracy (e.g., Friedrich et al., 2013). Since we did not rely on self-reports but on knowledge tests, we can conclude that the observed agreement between teachers' judgments and the SK tests was low and primarily stemmed from their shared association with school grades. However, we cannot determine which measure was more accurate. Nevertheless, we carefully designed all SK tests based on current knowledge and tested their reliability, factor structure, and validity in previous studies (Brunner et al., 2025; Karlen et al., 2026), whereas the teachers in our study reported mostly limited experience in diagnosing SRL. Another related concern is the potential mismatch between teachers' judgments of students' general SK versus the context-bound tasks in the SK tests. While teachers' judgments were not based on specific learning situations, the scenario-based tasks necessarily embedded items in fictitious learning situations to assess whether students know which strategies are useful in which learning situations. However, the difference in format between teachers' judgments and students' SK tests may limit direct comparability and reduce observed accuracy. Nevertheless, the SK tests included a wide variety of tasks, subject domains, and learning situations. This enabled the derivation of students' general SK from their performance across various tasks. Additionally, teachers were presented with example tasks to help them make informed judgments by understanding what kind of SK was assessed in the tests.

In future research, SRL could be assessed using a multimodal approach (de Mooij et al., 2025) to enhance validity. Technological advances enable tracking students' interactions with a learning system and the assessment of their SRL during the process (Azevedo, 2020). However, this focuses less on students' knowledge and more on their actual SRL behavior. Another option would be to use multiple teachers' judgments for the same student to estimate interrater reliability as an indicator of teachers' judgment consistency. Previous research showed that the agreement between teachers varied depending on the judged skill and was not always satisfactory (Muschkin et al., 2007).

The use of school grades as an indicator of academic achievement further complicates the issue. Students' school grades do not solely reflect academic competence but are influenced by various factors, such as personality (Hessen & Kuncel, 2022), behavior (DeVries et al., 2018), and SRL competencies (Richardson et al., 2012). This supports the interpretation of the achievement bias as a halo effect, with school grades closely aligned with teachers' overall impressions of students. Using a composite measure of school grades, such as

GPA, also aligns with this interpretation, as it may be a better indicator of teachers' overall impression of a student than a subject-specific grade. At the same time, a composite measure based on grades awarded by different teachers reduced the potential bias introduced by the same teachers giving SK judgments and assigning grades (Sprietsma, 2013).

Furthermore, it should be acknowledged that the tests assessing teachers' knowledge of SRL were newly developed for the current study and were not used or validated in prior studies. As with students' SK tests, teachers' tests included learning scenarios spanning multiple school subjects. It can be questioned whether teachers' knowledge of how to assess SRL strategies in a different school subject (e.g., Math for a literature teacher) is relevant to their judgment accuracy. Although many strategies are important across subjects (e.g., assessing prior knowledge), they may be more easily detected in a learning scenario that resembles teachers' own experiences. After all, subject-specific teaching contexts may offer distinct opportunities to observe strategies and provide individualized learning support. The absence of information on the subjects taught by participating teachers is thus considered a further limitation of the study. Nevertheless, assessing SK is important for teachers across all subjects. Future studies could explore how subject differences influence both teachers' SK judgments and their performance in subject-specific learning scenarios. Since SRL is a broad construct (Panadero, 2017), future studies should place particular emphasis on ensuring that the focus of the tests is consistent between teachers and students. An aspect not considered in the current study was teachers' pedagogical content knowledge for teaching SRL to meet students' needs (Karlen et al., 2024), which could also be relevant to their assessment.

Teachers' self-concept and diagnostic experience were assessed via self-reports, which are susceptible to bias. Research on self-assessment accuracy has shown that low performers tend to be overconfident and overestimate their performance, while high performers tend to be more accurate in their judgments and even tend to underestimate their performance (Hacker et al., 2014). Thus, teachers with low diagnostic skills may have overestimated their experience and reported a high self-concept, whereas those with high diagnostic skills tended to underestimate both. This could have contributed to the observed association between higher self-concept and experience with more biased judgments.

A further aspect is that teachers should collect information systematically over time and weigh it carefully to form an accurate judgment (Klug et al., 2013). In the study context, teachers were expected to judge their students' SK immediately while completing the questionnaire, independent of a specific observation. Therefore, they had to form a judgment without the opportunity to actively seek deep cues in subsequent lessons. This could have enhanced their reliance on easily observable surface cues. Moreover, their judgments depended on recalled observations, which may have been remembered incompletely and shaped by teachers' own beliefs and attributions (W. H. Cooper, 1981). In line with this, teachers primarily relied on achievement and gender when making memory-based judgments of students' motivation, but were able to consider diagnostic cues in other experimental conditions (Beck et al., 2025). Thus, the study context may have amplified the halo effect. However, qualitative research indicates that teachers rarely engage in systematic data collection and tend to rely heavily on intuition (Vanlommel et al., 2017), suggesting that the judgment process in the study may still reflect teachers' daily practice.

Taken together, our results highlight the need to better understand how teachers form their SK judgments. A promising approach might be to examine teachers' diagnostic reasoning (Kosel et al., 2024) or their think-aloud data (Glock et al., 2013). Furthermore, video-

based simulations of learning situations and teachers' respective diagnostic notes could provide additional insight into diagnostic processes (Radkowsch et al., 2023). However, these approaches capture only conscious processes and may even alter them by enhancing awareness and monitoring (Panadero et al., 2016). This could lead to an underestimation of unconscious biases, such as the halo effect. Consistent with this, Glock and colleagues (2013) found no indication of biased argumentation in think-aloud data, although students' ethnicity still influenced teachers' judgments. Eye-tracking data might indicate what teachers focus on when making their judgments and has been used in the context of teachers' judgments of the difficulty of a math task. There, a proportional focus on relevant task areas was associated with higher judgment accuracy (Brunner et al., 2024). Since the various research approaches address distinct aspects, combining them could provide a more comprehensive understanding of how teachers form their judgments.

## Conclusion

To summarize, the current study highlights teachers' persistent difficulties in judging students' SK, an important aspect of SRL. The observed judgment accuracy was low and could be attributed mainly to teachers' reliance on students' characteristics. This emphasizes the need for a cautious interpretation of relative accuracy, as the correlation coefficient does not directly indicate teachers' diagnostic competence. Teachers' judgments were affected by systematic achievement and gender biases, potentially driven by a halo effect largely independent of teachers' knowledge, experience, or self-concept. Considering these findings, we underscore the importance of enhancing teachers' awareness about judgment biases and supporting systematic, multi-perspective assessment of students' SRL through collaboration among multiple teachers, the use of rubrics, or digital diagnostic tools.

## Appendix A

### Examples of the developed learning traces and students' SK tests

Appendix A provides an example of the developed learning traces to assess teachers' assessment knowledge of SRL, the full scales used to assess teachers' diagnostic self-concept and gender stereotypes, and one example task per SK test. Some of these examples were also included in previous studies (Brunner et al., 2025; Karlen et al., 2026), which used the same SK tests. All items and tasks were translated from the study language into English.

### Learning traces to assess teachers' SRL assessment knowledge

Introduction text: The following will show how Nico approached a learning task. You will see a learning journal entry, which Nico completes both before and after studying. Additionally, you will be presented with Nico's notes on a geometry topic. Based on these two sources of information, assess Nico's self-regulated learning. Which strategies or aspects did he implement well, and which ones did he struggle with?



Learning Journal Entry (Date: Tuesday, 18.01.2022)

## Before Studying

What tasks do you have this week?

This week I have a sports exam in gymnastics. Since I really enjoy it, I'm already pretty good at everything. On Friday, I have a geometry test, and I really need a good grade in that one. Also, I should probably sort my study materials.

What are your goals for today?

I want to understand how to calculate the area of different shapes. I really need to know that for the test. I can sort my materials later.

How will you try to reach your goal?

I'll write down everything I already know and try to make drawings for the tricky formulas. That sometimes helps me.  
I'll write down everything I already know and try to make drawings for the tricky formulas. That sometimes helps me.

## After Studying

How much time did you spend studying today?

1.5 hours

What percentage of your goal did you reach today?

About 50%. I understand a few things better now. But not everything yet.

What went well while studying?

I was able to concentrate well. I gave my phone to my mom while I was studying. Otherwise, WhatsApp and TikTok distract me too much.  
Even though I could not find all my materials, I still learned a lot. A bit of mess in the room is normal anyway.

What was still hard?

I noticed that I still don't really understand how the formula works for triangles that aren't right-angled. I don't think I'll it out on my own. I'll probably ask my older brother.  
But knowing myself, it won't help much. Sometimes I just don't get it.

Is there anything you want to do better or differently tomorrow? If yes, what?

Tomorrow I'll ask my brother if he can help me with the triangles.

I already got a bad grade on the last geometry test. So, I really need a miracle.

**Area of triangles**

I already know how to calculate the area of a rectangle: height  $\times$  base. For a triangle, the formula is:

$$A = \frac{1}{2} \times b \times h$$

However, the height is not simply one of the triangle's sides; it must be drawn at a 90-degree angle. Except in the case of a right-angled triangle, where this is already correct.

**Right-angled triangle**

**Other triangle**

Here I understand the formula: you can form a rectangle that is twice as large, which is why you have to divide by two.

Why is it not allowed to simply use the side here? I don't understand that.

Fig. 2 Learning journal entry and materials with learning traces - example Nico

Table 5 Coding examples for the evaluation of Nico's SRL

Examples of...

... predefined correct answers

... teachers' answers considered correct

... teachers' answers considered incorrect

well-executed strategies

Creating visual representations (cognitive strategies)

"Visualization of mathematical formulas"

Reflecting on what is understood and what is not (metacognitive strategy)

"Taking prior knowledge into account"

"Analysis of one's own knowledge"

Setting priorities: addressing geometry as the primary task (time management strategy)

"Prioritization: Focusing on mathematics learning"

"summary"

poorly executed strategies

Disorganized workspace (learning environment management)

"Organization of his personal study materials"

"keywords instead of full sentences"

Growth mindset is missing ("sometimes I just don't get it")

"Self-confidence has not been reinforced"

"I would not write down information I already know."

Coping with failure: belief that a previous unsuccessful exam predicts future failure.

"procrastination"

These are only some examples; there were more predefined correct answers. Teachers' answers were translated into English



## Teachers' diagnostic self-concept regarding SRL

**Table 6** Items to assess teachers' diagnostic self-concept regarding SRL

| To what extent do the following statements apply to you?                                        |                                                                                  |
|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Item 1                                                                                          | I can judge students' competencies in self-regulated learning well.              |
| Item 2                                                                                          | I find it easy to identify where students struggle with self-regulated learning. |
| Item 3                                                                                          | I am good at assessing students' competencies in self-regulated learning.        |
| Items were rated on a six-point scale (1 = "Does not apply at all" to 6 = "applies completely") |                                                                                  |

## Teachers' gender stereotypes regarding SRL

**Table 7** Items to assess teachers' gender stereotypes regarding SRL

| Are there differences in self-regulated learning between girls and boys?                                                                           |                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Item 1                                                                                                                                             | Who is better able to take responsibility for their own learning? |
| Item 2                                                                                                                                             | Who copes better in open learning environments?                   |
| Item 3                                                                                                                                             | Who can work more independently on long-term projects?            |
| Items were rated on a scale from 1–5 (1 = "boys much better", 2 = "boys better", 3 = "both the same", 4 = "girls better", 5 = "girls much better") |                                                                   |

## Example tasks for the SK tests

**Table 8** Example task for SK of cognitive strategies

| Tom has to prepare for an exam on viruses and bacteria. He has devised strategies for the individual learning tasks. Which strategy is best suited to which learning task? Arrange the different learning strategies on the right side to match the learning tasks on the left side. |                                                                              |                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------|
|                                                                                                                                                                                                                                                                                      | Learning task                                                                | Strategies                     |
| Subtask 1                                                                                                                                                                                                                                                                            | Understand how a virus is structured.                                        | He writes index cards.         |
| Subtask 2                                                                                                                                                                                                                                                                            | Know the 20 most important bacteria and viruses and the diseases they cause. | He writes a summary.           |
| Subtask 3                                                                                                                                                                                                                                                                            | Know the five most important differences between bacteria and viruses.       | He thinks up mnemonic bridges. |
| Subtask 4                                                                                                                                                                                                                                                                            | Understand why vaccinations and antibiotics are so important in medicine.    | He draws a picture.            |
| Each correctly solved subtask was awarded 1 point, with the mean score calculated to receive a task score ranging from 0 to 1                                                                                                                                                        |                                                                              |                                |

**Table 9** Example task for SK of metacognitive strategies

| Even though he had studied, Elias got a bad grade on a mock exam. He thinks about how to better prepare for the real exam this time. Rate how good the individual ideas are! (1 = not good at all, 4 = very good) |                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Subtasks                                                                                                                                                                                                          | Strategies                                                                                                     |
| a                                                                                                                                                                                                                 | He tries not to think about the bad mock exam.                                                                 |
| b                                                                                                                                                                                                                 | He thinks about what he can do better in the real exam.                                                        |
| c                                                                                                                                                                                                                 | He makes flashcards and memorizes the solutions of the mock exam.                                              |
| d                                                                                                                                                                                                                 | He thinks about what he did wrong in the mock exam and practices again the tasks he could not solve correctly. |
| e                                                                                                                                                                                                                 | He solves all tasks again and again.                                                                           |
| f                                                                                                                                                                                                                 | He plans to observe himself during the real exam and correct mistakes directly.                                |
| g                                                                                                                                                                                                                 | He resolves not to make a single mistake in the real exam.                                                     |

For each pair comparison (e.g., strategy a > strategy b), zero or one point was awarded, depending on whether students' ratings aligned with the experts' ratings. The mean of all pair comparisons constituted the task score, ranging from 0 to 1

**Table 10** Example task for SK of time management

| Ivana has several exams scheduled for next week. She is wondering how to study most effectively to be well prepared for all of them. Choose the better option from each of the three tips presented |                                                                                                    |                                                                         |                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------|
| Subtasks                                                                                                                                                                                            | Tip 1                                                                                              | Tip 2                                                                   | Tip 3                                                            |
| a                                                                                                                                                                                                   | She should study for extended periods and then reward herself with a long break.                   | She should interrupt her study sessions occasionally with short breaks. | She should not take breaks and use all her time for studying.    |
| b                                                                                                                                                                                                   | She should schedule short study sessions for each subject and distribute them throughout the week. | She should leave out specific subjects to make better use of her time.  | She should schedule two complete study days for each subject.    |
| c                                                                                                                                                                                                   | She should learn the words the evening before the exam.                                            | She should learn the words in a single day.                             | She should study the words in smaller blocks over multiple days. |
| d                                                                                                                                                                                                   | She should always study in the morning before school.                                              | She should study whenever she feels like it.                            | She should develop a habit of studying at fixed times.           |

For each correct subtask, one point was awarded; otherwise, none. To receive a task score ranging from 0 to 1, the mean score of all subtask scores was calculated

**Table 11** Example task for SK of management of the learning environment

Shanti struggles to learn independently outside school. To find out how she can learn better, she tries out different options. Choose the best of the three options.

Option 1:

Shanti learns during the bus ride to school.

Option 2:

Shanti studies on her bed because she is tired.

Option 3:

Shanti studies at the living room table. That is where she has the most space.

For the correct choice, one point was awarded

**Table 12** Example task for SK of motivation regulation

Pia is at home and should be studying for a biology test. However, the subject is difficult, and she thinks she will not pass the test anyway. She has a few ideas on how she could motivate herself. Rate how good each idea is. (1 = not good at all, 4 = very good).

Subtasks

Strategies

a

She chooses a time for learning when she can concentrate well.

b

She tells herself that she passed the last difficult test and can do it again this time.

c

If she is unmotivated, learning is useless, so she puts it off until the next day.

d

She thinks about meeting with her friends later and quickly skims the chapters in her textbook.

e

She thinks about how embarrassing a bad grade in biology would be.

f

She tells herself she will meet up with her friends to study. If they explained the content to each other, she surely would pass the test.

For each pair comparison (e.g., strategy b > strategy c), zero or one point was awarded, depending on whether students' ratings aligned with the experts' ratings. The mean of all pair comparisons constituted the task score, ranging from 0 to 1

## Appendix B

### Results of additional analyses

**Table 13** Nesting of teacher judgments and student SK in classes

| SRL components                  | Teacher judgments |        |       |       | Student SK |        |       |        |
|---------------------------------|-------------------|--------|-------|-------|------------|--------|-------|--------|
|                                 | ANOVA             |        |       |       | ANOVA      |        |       |        |
|                                 | ICC(1)            | ICC(2) | F     | p     | ICC(1)     | ICC(2) | F     | p      |
| Cognitive strategies            | 0.214             | 0.441  | 1.790 | 0.002 | 0.221      | 0.433  | 1.764 | 0.004  |
| Metacognitive strategies        | 0.106             | 0.256  | 1.343 | 0.076 | 0.228      | 0.456  | 1.839 | 0.002  |
| Time management strategies      | 0.141             | 0.323  | 1.476 | 0.029 | 0.258      | 0.498  | 1.991 | <0.001 |
| Learning environment strategies | 0.149             | 0.337  | 1.509 | 0.023 | 0.093      | 0.230  | 1.299 | 0.104  |
| Motivation regulation           | 0.069             | 0.178  | 1.216 | 0.170 | 0.032      | 0.086  | 1.094 | 0.329  |

ICC(1) indicates the proportion of total variance attributable to differences between clusters, while ICC(2) reflects the proportion of total variance that could be explained if only the class means were used

**Table 14** Results of model 2 using predictors centered within clusters

|                       | Cognitive strategies      |          | Metacognitive strategies  |          | Time management           |          | Learning environment      |          | Motivation regulation     |          |
|-----------------------|---------------------------|----------|---------------------------|----------|---------------------------|----------|---------------------------|----------|---------------------------|----------|
|                       | <i>B</i> ( <i>SE</i> )    | <i>p</i> | <i>B</i> ( <i>SE</i> )    | <i>p</i> | <i>B</i> ( <i>SE</i> )    | <i>p</i> | <i>B</i> ( <i>SE</i> )    | <i>p</i> | <i>B</i> ( <i>SE</i> )    | <i>p</i> |
| <i>Fixed effects</i>  |                           |          |                           |          |                           |          |                           |          |                           |          |
| Intercept             | 2.757<br>(0.075)          | <0.001   | 2.802<br>(0.069)          | <0.001   | 3.090<br>(0.080)          | <0.001   | 3.535<br>(0.080)          | <0.001   | 2.909<br>(0.074)          | <0.001   |
| SK                    | 0.002<br>(0.004)          | 0.569    | 0.005<br>(0.003)          | 0.111    | 0.005<br>(0.004)          | 0.172    | 0.002<br>(0.003)          | 0.431    | 0.001<br>(0.003)          | 0.742    |
| Grades                | 0.824<br>(0.094)          | <0.001   | 0.859<br>(0.092)          | <0.001   | 0.728<br>(0.088)          | <0.001   | 0.526<br>(0.089)          | <0.001   | 0.787<br>(0.097)          | <0.001   |
| Gender                | -0.032<br>(0.099)         | 0.744    | -0.171<br>(0.095)         | 0.072    | -0.204<br>(0.106)         | 0.055    | -0.553<br>(0.102)         | <0.001   | -0.295<br>(0.104)         | 0.005    |
| SES                   | -0.029<br>(0.033)         | 0.375    | 0.020<br>(0.031)          | 0.515    | 0.011<br>(0.034)          | 0.754    | -0.014<br>(0.034)         | 0.675    | 0.028<br>(0.034)          | 0.416    |
| <i>Random effects</i> | Variance<br>( <i>SD</i> ) |          | Variance<br>( <i>SD</i> ) |          | Variance<br>( <i>SD</i> ) |          | Variance<br>( <i>SD</i> ) |          | Variance<br>( <i>SD</i> ) |          |
| Intercept             | 0.061<br>(0.247)          |          | 0.027<br>(0.164)          |          | 0.090<br>(0.299)          |          | 0.104<br>(0.322)          |          | 0.018<br>(0.134)          |          |
| Residuals             | 0.310<br>(0.556)          |          | 0.321<br>(0.567)          |          | 0.353<br>(0.594)          |          | 0.341<br>(0.584)          |          | 0.400<br>(0.633)          |          |

**Table 15** Models with cross-level interactions for teachers' judgments of time management

| <i>Model</i>               | Model 2                | Interaction Models     |                         |                        |
|----------------------------|------------------------|------------------------|-------------------------|------------------------|
|                            |                        | Diagnostic experience  | Diagnostic self-concept | Both                   |
| <i>n</i> level 1           | 164                    | 164                    | 164                     | 164                    |
| <i>n</i> level 2           | 59                     | 59                     | 59                      | 59                     |
| <i>Fixed effects</i>       | <i>B</i> ( <i>SE</i> ) | <i>B</i> ( <i>SE</i> ) | <i>B</i> ( <i>SE</i> )  | <i>B</i> ( <i>SE</i> ) |
| Intercept                  | 3.080 (0.081)          | 3.077 (0.079)          | 3.075 (0.080)           | 3.076 (0.079)          |
| SK                         | 0.084 (0.057)          | 0.097 (0.056)          | 0.080 (0.057)           | 0.093 (0.057)          |
| Grades                     | 0.445 (0.056)***       | 0.453 (0.055)***       | 0.467 (0.056)***        | 0.460 (0.056)***       |
| Gender                     | -0.200 (0.106)         | -0.187 (0.104)         | -0.191 (0.105)          | -0.186 (0.104)         |
| SES                        | 0.011 (0.054)          | 0.012 (0.053)          | 0.002 (0.053)           | 0.008 (0.053)          |
| <i>Interactions</i>        |                        |                        |                         |                        |
| Gender x Dia Exp           |                        | -0.271 (0.105)*        |                         | -0.217 (0.135)         |
| Gender x Dia SC            |                        |                        | -0.244 (0.116)*         | -0.093 (0.148)         |
| <i>Random effects</i>      | Variance ( <i>SD</i> ) | Variance ( <i>SD</i> ) | Variance ( <i>SD</i> )  | Variance ( <i>SD</i> ) |
| Intercept                  | 0.088 (0.297)          | 0.082 (0.286)          | 0.087 (0.294)           | 0.083 (0.288)          |
| Residual                   | 0.353 (0.564)          | 0.341 (0.584)          | 0.343 (0.586)           | 0.339 (0.583)          |
| <i>Model fit</i>           |                        |                        |                         |                        |
| AIC                        | 338.831                | 334.294                | 336.464                 | 335.898                |
| BIC                        | 360.530                | 359.093                | 361.263                 | 363.797                |
| R <sup>2</sup> conditional | 0.500                  | 0.516                  | 0.512                   | 0.518                  |
| R <sup>2</sup> marginal    | 0.375                  | 0.400                  | 0.389                   | 0.401                  |
| ICC adjusted               | 0.199                  | 0.193                  | 0.202                   | 0.196                  |

*Dia Exp* Diagnostic Experience, *Dia SC* Diagnostic self-concept. Student-level predictors were z-transformed, and teacher-level predictors were grand-mean centered. Results indicate the best fit for the model, which includes the interaction between gender and diagnostic experience, whereby the significance of the difference could not be tested due to the varying predictors included in the models. If both interactions were included in one model, neither of them was significant, indicating shared variance between both teacher variables

**Acknowledgements** We thank everyone included in the data collection and project implementation, especially Prof. Dr. Joachim Wirth (Ruhr University of Bochum), Xenia-Lea Weber-Reuter (Ruhr University Bochum), Prof. Dr. Ferdinand Stebner (University of Osnabrück), Dr. Kerstin Bäuerlein (University of Applied Sciences and Arts Northwestern Switzerland), and Martina Conti (St. Gallen University of Teacher Education).

**Authors' contributions** Sabrina Nadja Brunner; Conceptualization, data curation, Formal analysis, Writing – original draft, Writing –review and editing Yves Karlen; Conceptualization, Funding acquisition, Project administration, Supervision, Writing – original draft, Writing– review and editing.

**Funding** The authors acknowledge financial support for this research from the Robert Bosch Institute (01000795-001).

**Data availability** Data and code can be provided upon request.

## Declarations

**Ethical approval** Was granted by the Ethics Committee, and written consent was obtained from the legal guardians of the participating students, who were free to withdraw from the study at any time.

**Conflicts of interest** The authors declare declare that they have no conflicts of interest.

**Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

## References

- Adams, K. S., & Christenson, S. L. (2000). Trust and the family–school relationship examination of parent–teacher differences in elementary and secondary grades. *Journal of School Psychology, 38*(5), 477–497. [https://doi.org/10.1016/S0022-4405\(00\)00048-0](https://doi.org/10.1016/S0022-4405(00)00048-0)
- Adams, R. V., & Blair, E. (2019). Impact of time management behaviors on undergraduate engineering students' performance. *Sage Open, 9*(1), 1–11. <https://doi.org/10.1177/2158244018824506>
- Azevedo, R. (2020). Reflections on the field of metacognition: Issues, challenges, and opportunities. *Metacognition and Learning, 15*(2), 91–98. <https://doi.org/10.1007/s11409-020-09231-x>
- Bäuerlein, K., Brunner, S.N., Etter, B., Stebner, F., Weber, X.-L., & Karlen, Y. (2023). Förderung der diagnostischen Kompetenz von Lehrpersonen im selbstregulierten Lernen.[Fostering teachers' diagnostic competence in self-regulated learning] In R. Porsch & P. Gollub (Eds.), *Professionalisierung im Lehrerberuf [Professionalization in the Teaching Profession]* (pp. 159–177). Waxmann. <https://doi.org/10.31244/9783830997429>
- Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software, 67*(1), 1–48. <https://doi.org/10.18637/jss.v067.i01>
- Bechger, T. M., Maris, G., & Hsiao, Y. P. (2010). Detecting halo effects in performance-based examinations. *Applied Psychological Measurement, 34*(8), 607–619. <https://doi.org/10.1177/0146621610367897>
- Beck, J., Dutke, S., & Utesch, T. (2025). Understanding teacher judgments of student motivation: The role of (un-)available cues. *Learning and Instruction, 95*, Article 102029. <https://doi.org/10.1016/j.learninstruc.2024.102029>

- Behrmann, L. (2019). The halo effect as a teaching tool for fostering research-based learning. *European Journal of Educational Research*, 8(2), 433–441. <https://doi.org/10.12973/eu-jer.8.2.433>
- Bergold, S., & Steinmayr, R. (2023). Teacher judgments predict developments in adolescents' school performance, motivation, and life satisfaction. *Journal of Educational Psychology*, 115(4), 642–664. <https://doi.org/10.1037/edu0000786>
- Bidjerano, T. (2005). *Gender differences in self-regulated learning*. Annual Meeting of the Northeastern Educational Research Association, Kerhonkson, NY.
- Biernat, M., Manis, M., & Nelson, T. E. (1991). Stereotypes and standards of judgment. *Journal of Personality and Social Psychology*, 60(4), 485–499. <https://doi.org/10.1037/0022-3514.60.4.485>
- Boekaerts, M. (1997). Self-regulated learning: A new concept embraced by researchers, policy makers, educators, teachers, and students. *Learning and Instruction*, 7(2), 161–186. [https://doi.org/10.1016/S0959-4752\(96\)00015-1](https://doi.org/10.1016/S0959-4752(96)00015-1)
- Bong, M., & Skaalvik, E. M. (2003). Academic self-concept and self-efficacy: How different are they really? *Educational Psychology Review*, 15(1), 1–40. <https://doi.org/10.1023/A:1021302408382>
- Brunner, K., Obersteiner, A., & Leuders, T. (2024). How pedagogical content knowledge sharpens prospective teachers' focus when judging mathematical tasks: An eye-tracking study. *Educational Studies in Mathematics*, 115(2), 177–196. <https://doi.org/10.1007/s10649-023-10281-6>
- Brunner, S. N., Bäuerlein, K., Conti, M., & Karlen, Y. (2025). Measuring individual differences in students' knowledge about self-regulated learning strategies with a digital tool. *Learning and Individual Differences*, 119, 102656. <https://doi.org/10.1016/j.lindif.2025.102656>
- Callan, G. L., & Shim, S. S. (2019). How teachers define and identify self-regulated learning. *The Teacher Educator*, 54(3), 295–312. <https://doi.org/10.1080/08878730.2019.1609640>
- Caro, D. H., McDonald, J. T., & Willms, J. D. (2009). Socio-economic status and academic achievement trajectories from childhood to adolescence. *Canadian Journal of Education / Revue Canadienne de l'éducation*, 32(3), 558–590.
- Carr, M., & Kurtz, B. E. (1991). Teachers' perceptions of their students' metacognition, attributions, and self-concept. *British Journal of Educational Psychology*, 61(2), 197–206. <https://doi.org/10.1111/j.2044-8279.1991.tb00975.x>
- Clift, R. T., Ghatala, E. S., Naus, M. M., & Poole, J. (1990). Exploring teachers' knowledge of strategic study activity. *The Journal of Experimental Education*, 58(4), 253–263. <https://doi.org/10.1080/00220973.1990.10806539>
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Routledge. <https://doi.org/10.4324/9780203771587>
- Cole, D. A., Martin, J. M., Peeke, L. A., Seroczynski, A. D., & Fier, J. (1999). Children's over- and underestimation of academic competence: A longitudinal study of gender differences, depression, and anxiety. *Child Development*, 70(2), 459–473. <https://doi.org/10.1111/1467-8624.00033>
- Cooper, H. M. (1979). Pygmalion grows up: A model for teacher expectation communication and performance influence. *Review of Educational Research*, 49(3), 389–410. <https://doi.org/10.3102/003465430490033>
- Cooper, W. H. (1981). Ubiquitous halo. *Psychological Bulletin*, 90(2), 218–244. <https://doi.org/10.1037/0033-2909.90.2.218>
- Corno, L. (2008). On teaching adaptively. *Educational Psychologist*, 43(3), 161–173. <https://doi.org/10.1080/00461520802178466>
- Cronbach, L. J. (1951). Coefficient alpha and the internal structure of the tests. *Psychometrika*, 16(3), 297–334. <https://doi.org/10.1007/BF02310555>
- de Boer, H., Donker, A. S., Kostons, D. D. N. M., & van der Werf, G. P. C. (2018). Long-term effects of meta-cognitive strategy instruction on student academic performance: A meta-analysis. *Educational Research Review*, 24, 98–115. <https://doi.org/10.1016/j.edurev.2018.03.002>
- de Bruin, A. B. H., Roelle, J., Carpenter, S. K., Baars, M., EFG-MRE. (2020). Synthesizing cognitive load and self-regulation theory: A theoretical framework and research agenda. *Educational Psychology Review*, 32(4), 903–915. <https://doi.org/10.1007/s10648-020-09576-4>
- de Mooij, S., Lämsä, J., Lim, L., Aksela, O., Athavale, S., Bistolfi, I., Jin, F., Li, T., Azevedo, R., Bannert, M., Gašević, D., Järvelä, S., & Molenaar, I. (2025). A systematic review of self-regulated learning through integration of multimodal data and artificial intelligence. *Educational Psychology Review*, 37(2), Article 54. <https://doi.org/10.1007/s10648-025-10028-0>
- Dersch, A.-S., Heyder, A., & Eitel, A. (2022). Exploring the nature of teachers' math-gender stereotypes: The Math-Gender Misconception Questionnaire. *Frontiers in Psychology*, 13, Article 820254. <https://doi.org/10.3389/fpsyg.2022.820254>
- DeVries, J. M., Rathmann, K., & Gebhardt, M. (2018). How does social behavior relate to both grades and achievement scores? *Frontiers in Psychology*, 9, Article 857. <https://doi.org/10.3389/fpsyg.2018.00857>

- Di Battista, S., Pivetti, M., & Bozzi, G. (2024). Brief research report: Teachers' gender-differentiated attributions. *The Journal of Experimental Education*, 92(1), 120–129. <https://doi.org/10.1080/00220973.2022.2141175>
- Dignath, C., & Sprenger, L. (2020). Can you only diagnose what you know? The relation between teachers' self-regulation of learning concepts and their assessment of students' self-regulation. *Frontiers in Education*, 5, Article 585683. <https://doi.org/10.3389/feduc.2020.585683>
- Dompnier, B., Pansu, P., & Bressoux, P. (2006). An integrative model of scholastic judgments: Pupils' characteristics, class context, halo effect and internal attributions. *European Journal of Psychology of Education*, 21(2), 119–133. <https://doi.org/10.1007/BF03173572>
- Dörrenbächer-Ulrich, L., Sparfeldt, J. R., & Perels, F. (2024). Knowing how to learn: Development and validation of the strategy knowledge test for self-regulated learning (SKT-SRL) for college students. *Metacognition and Learning*, 19, 1–45. <https://doi.org/10.1007/s11409-024-09379-w>
- Dörrenbächer-Ulrich, L., Weißenfels, M., Russer, L., & Perels, F. (2021). Multimethod assessment of self-regulated learning in college students: Different methods for different components? *Instructional Science*, 49(1), 137–163. <https://doi.org/10.1007/s11251-020-09533-2>
- Dresel, M., Schmitz, B., Schober, B., Spiel, C., Ziegler, A., Engelschalk, T., Jöstl, G., Klug, J., Roth, A., Wimmer, B., & Steuer, G. (2015). Competencies for successful self-regulated learning in higher education: Structural model and indications drawn from expert interviews. *Studies in Higher Education*, 40(3), 454–470. <https://doi.org/10.1080/03075079.2015.1004236>
- Duckworth, A. L., Gendler, T. S., & Gross, J. J. (2014). Self-control in school-age children. *Educational Psychologist*, 49(3), 199–217. <https://doi.org/10.1080/00461520.2014.926225>
- Edelsbrunner, P. A., Simonsmeier, B. A., & Schneider, M. (2025a). The Cronbach's alpha of domain-specific knowledge tests before and after learning: A meta-analysis of published studies. *Educational Psychology Review*, 37(1), 4. <https://doi.org/10.1007/s10648-024-09982-y>
- Edelsbrunner, P. A., Simonsmeier, B. A., & Schneider, M. (2025b). The reliability, but not the Cronbach's alpha, of knowledge tests matters: Response to Zitzmann and Orona (2025b). *Educational Psychology Review*, 37(2), Article 48. <https://doi.org/10.1007/s10648-025-10023-5>
- Efklides, A., & Schwartz, B. L. (2024). Revisiting the metacognitive and affective model of self-regulated learning: Origins, development, and future directions. *Educational Psychology Review*, 36(2), 61. <https://doi.org/10.1007/s10648-024-09896-9>
- Enders, C. K., & Tofighi, D. (2007). Centering predictor variables in cross-sectional multilevel models: A new look at an old issue. *Psychological Methods*, 12(2), 121–138. <https://doi.org/10.1037/1082-989X.12.2.121>
- Esarey, J., & Sumner, J. L. (2017). Marginal effects in interaction models: Determining and controlling the false positive rate. *Comparative Political Studies*, 51(9), 1144–1176. <https://doi.org/10.1177/0010414017730080>
- Evans, J. S. B. T. (2010). Intuition and reasoning: A dual-process perspective. *Psychological Inquiry*, 21(4), 313–326. <https://doi.org/10.1080/1047840X.2010.521057>
- Flavell, J. H. (1979). Metacognition and cognitive monitoring. A new area of cognitive-developmental inquiry. *American Psychologist*, 34(10), 906–911. <https://doi.org/10.1037/0003-066X.34.10.906>
- Flippo, R. F., & Caverly, D. C. (Eds.). (2009). *Handbook of college reading and study strategy research* (2nd ed.). Routledge.
- Fox, J., & Monette, G. (1992). Generalized collinearity diagnostics. *Journal of the American Statistical Association*, 87(417), 178–183. <https://doi.org/10.1080/01621459.1992.10475190>
- Friedrich, A., Jonkmann, K., Nagengast, B., Schmitz, B., & Trautwein, U. (2013). Teachers' and students' perceptions of self-regulated learning and math competence: Differentiation and agreement. *Learning and Individual Differences*, 27, 26–34. <https://doi.org/10.1016/j.lindif.2013.06.005>
- Funder, D. C. (1995). On the accuracy of personality judgment: A realistic approach. *Psychological Review*, 102(4), 652–670. <https://doi.org/10.1037/0033-295X.102.4.652>
- Glock, S., Krolak-Schwerdt, S., Klapproth, F., & Böhmer, M. (2013). Beyond judgment bias: How students' ethnicity and academic profile consistency influence teachers' tracking judgments. *Social Psychology of Education*, 16(4), 555–573. <https://doi.org/10.1007/s11218-013-9227-5>
- Greene, J. A., Bolick, C. M., Jackson, W. P., Caprino, A. M., Oswald, C., & McVea, M. (2015). Domain-specificity of self-regulated learning processing in science and history. *Contemporary Educational Psychology*, 42, 111–128. <https://doi.org/10.1016/j.cedpsych.2015.06.001>
- Hacker, D. J., Bol, L., & Keener, M. C. (2014). Metacognition in education. *Handbook of Metamemory and Memory*. Routledge. <https://doi.org/10.4324/9780203805503.ch22>
- Händel, M., Artelt, C., & Weinert, S. (2013). Assessing metacognitive knowledge: Development and evaluation of a test instrument. *Journal for Educational Research Online*, 5(2), 162–188. <https://doi.org/10.25656/01:8429>

- Händel, M., Lockl, K., Heydrich, J., Weinert, S., & Artelt, C. (2014). Assessment of metacognitive knowledge in students with special educational needs. *Metacognition and Learning*, 9(3), 333–352. <https://doi.org/10.1007/s11409-014-9119-x>
- Harris, M. J., & Rosenthal, R. (1985). Mediation of interpersonal expectancy effects: 31 meta-analyses. *Psychological Bulletin*, 97(3), 363–386. <https://doi.org/10.1037/0033-2909.97.3.363>
- Heppt, B., Olczyk, M., & Volodina, A. (2022). Number of books at home as an indicator of socioeconomic status: Examining its extensions and their incremental validity for academic achievement. *Social Psychology of Education*, 25(4), 903–928. <https://doi.org/10.1007/s11218-022-09704-8>
- Hessen, P. R., & Kuncel, N. R. (2022). Beyond grades: A meta-analysis of personality predictors of academic behavior in middle school and high school. *Personality and Individual Differences*, 199, Article 111809. <https://doi.org/10.1016/j.paid.2022.111809>
- Hu, L., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling: A Multidisciplinary Journal*, 6(1), 1–55. <https://doi.org/10.1080/10705519909540118>
- Jaeger, B. (2025). *r2glmm: Computes R squared for mixed (multilevel) models* (Version 0.1.3) [R package]. <https://CRAN.R-project.org/package=r2glmm>
- Jansen, T., Vögelin, C., Machts, N., Keller, S. D., & Möller, J. (2021). Don't just judge the spelling! The influence of spelling on assessing second-language student essays. *Frontline Learning Research*, 9(1), 44–65. <https://doi.org/10.14786/flr.v9i1.541>
- Jeong, S., & Feldon, D. F. (2023). Changes in self-regulated learning profiles during an undergraduate peer-based intervention: A latent profile transition analysis. *Learning and Instruction*, 83, Article 101710. <https://doi.org/10.1016/j.learninstruc.2022.101710>
- Jönsson, A., & Panadero, E. (2017). The use and design of rubrics to support assessment for learning. In D. Carless, S. M. Bridges, C. K. Y. Chan, & R. Glofcheski (Eds.), *Scaling up assessment for learning in higher education* (Vol. 5, pp. 99–111). Springer. [https://doi.org/10.1007/978-981-10-3045-1\\_7](https://doi.org/10.1007/978-981-10-3045-1_7)
- Kaiser, J., Retelsdorf, J., Südkamp, A., & Möller, J. (2013). Achievement and engagement: How student characteristics influence teacher judgments. *Learning and Instruction*, 28, 73–84. <https://doi.org/10.1016/j.learninstruc.2013.06.001>
- Kalkbrenner, M. T. (2023). Alpha, omega, and *H* internal consistency reliability estimates: Reviewing these options and when to use them. *Counseling Outcome Research and Evaluation*, 14(1), 77–88. <https://doi.org/10.1080/21501378.2021.1940118>
- Karing, C., Rausch, T., & Artelt, C. (2024). Teacher judgement accuracy—Measurements, causes and effects. In S. Weinert, H.-G. Rossbach, J. von Maurice, H.-P. Blossfeld, & C. Artelt (Eds.), *Educational processes, decisions, and the development of competencies from early preschool age to adolescence* (Vol. 16, pp. 263–280). Springer Fachmedien. [https://doi.org/10.1007/978-3-658-43414-4\\_10](https://doi.org/10.1007/978-3-658-43414-4_10)
- Karlen, Y. (2016a). Differences in students' metacognitive strategy knowledge, motivation, and strategy use: A typology of self-regulated learners. *The Journal of Educational Research*(3). <https://doi.org/10.1080/00220671.2014.942895>
- Karlen, Y. (2016b). Perceived learning environments and metacognitive strategy knowledge at the upper secondary school level. *Journal for Educational Research Online*, 8(2), 212–232. <https://doi.org/10.25656/01:12442>
- Karlen, Y., Bäuerlein, K., & Brunner, S.N. (2023). Teachers' assessment of self-regulated learning: Linking professional competences, assessment practices, and judgment accuracy. *Social Psychology of Education*, 27, 461–491. <https://doi.org/10.1007/s11218-023-09845-4>
- Karlen, Y., Brunner, S.N., Stebner, F., Conti, M., Bäuerlein, K. (2026). Making self-regulated learning visible: A digital tool for assessing and promoting self-regulated learning. [Manuscript submitted for publication]. Institute of education, University of Zurich.
- Karlen, Y., & Hertel, S. (2024). Inspiring self-regulated learning in everyday classrooms: Teachers' professional competences and promotion of self-regulated learning. *Unterrichtswissenschaft*, 52, 1–13. <https://doi.org/10.1007/s42010-024-00196-3>
- Karlen, Y., Hertel, S., Grob, U., Jud, J., & Hirt, C. N. (2024). Teachers matter: Linking teachers and students' self-regulated learning. *Research Papers in Education*, 40(3), 414–441. <https://doi.org/10.1080/02671522.2024.2394059>
- Karlen, Y., Hertel, S., & Hirt, C. N. (2020). Teachers' professional competences in self-regulated learning: An approach to integrate teachers' competences as self-regulated learners and as agents of self-regulated learning in a holistic manner. *Frontiers in Education*, 5, Article 159. <https://doi.org/10.3389/educ.2020.00159>
- Karlen, Y., Hirt, C. N., Jud, J., Rosenthal, A., & Eberli, T. D. (2023). Teachers as learners and agents of self-regulated learning: The importance of different teachers competence aspects for promoting metacognition. *Teaching and Teacher Education*, 125. <https://doi.org/10.1016/j.tate.2023.104055>



- Karlen, Y., Merki, K. M., & Ramseier, E. (2014). The effect of individual differences in the development of metacognitive strategy knowledge. *Instructional Science*, 42(5), 777–794. <https://doi.org/10.1007/s11251-014-9314-9>
- Kelty, N. E., & Wakabayashi, T. (2020). Family engagement in schools: Parent, educator, and community perspectives. *Sage Open*, 10(4), Article 2158244020973024. <https://doi.org/10.1177/2158244020973024>
- Kirschner, P. A., & Stoyanov, S. (2020). Educating youth for nonexistent/not yet existing professions. *Educational Policy*, 34(3), 477–517. <https://doi.org/10.1177/0895904818802086>
- Klug, J., Bruder, S., Kelava, A., & Schmitz, B. (2013). Diagnostic competence of teachers: A process model that accounts for diagnosing learning behavior tested by means of a case scenario. *Teaching and Teacher Education*, 30, 38–46. <https://doi.org/10.1016/j.tate.2012.10.004>
- Klug, J., Bruder, S., & Schmitz, B. (2016). Which variables predict teachers diagnostic competence when diagnosing students' learning behavior at different stages of a teacher's career? *Teachers and Teaching*, 22(4), 461–484. <https://doi.org/10.1080/13540602.2015.1082729>
- Koller, M. (2016). robustlmm: An R package for robust estimation of linear mixed-effects models. *Journal of Statistical Software*, 75(6). <https://doi.org/10.18637/jss.v075.i06>
- Kolovou, D., Hochweber, J., & Praetorius, A.-K. (2024). Does teacher judgment accuracy matter? How judgment accuracy, teaching quality, and student achievement development are related. *Teaching and Teacher Education*, 144, Article 104555. <https://doi.org/10.1016/j.tate.2024.104555>
- Koriat, A. (1997). Monitoring one's own knowledge during study: A cue-utilization approach to judgments of learning. *Journal of Experimental Psychology: General*, 126(4), 349–370. <https://doi.org/10.1037/0096-3445.126.4.349>
- Kosel, C., Bauer, E., & Seidel, T. (2024). Where experience makes a difference: Teachers' judgment accuracy and diagnostic reasoning regarding student learning characteristics. *Frontiers in Psychology*, 15, Article 1278472. <https://doi.org/10.3389/fpsyg.2024.1278472>
- Latimier, A., Peyre, H., & Ramus, F. (2021). A meta-analytic review of the benefit of spacing out retrieval practice episodes on retention. *Educational Psychology Review*, 33(3), 959–987. <https://doi.org/10.1007/s10648-020-09572-8>
- Liu, X., He, W., Zhao, L., & Hong, J.-C. (2021). Gender differences in self-regulated online learning during the COVID-19 lockdown. *Frontiers in Psychology*, 12, Article 752131. <https://doi.org/10.3389/fpsyg.2021.752131>
- Loibl, K., Leuders, T., & Dörfler, T. (2020). A framework for explaining teachers' diagnostic judgements by cognitive modeling (DiaCoM). *Teaching and Teacher Education*, 91, Article 103059. <https://doi.org/10.1016/j.tate.2020.103059>
- Long, J. A. (2024). *Interactions: Comprehensive, user-friendly toolkit for probing interactions* (Version R package version 1.2.0) [Computer software]. <https://doi.org/10.32614/CRAN.package.interactions>
- Lovakov, A., & Agadullina, E. R. (2021). Empirically derived guidelines for effect size interpretation in social psychology. *European Journal of Social Psychology*, 51(3), 485–504. <https://doi.org/10.1002/ejsp.2752>
- Lüdtke, D., Ben-Shachar, M., Patil, I., Waggoner, P., & Makowski, D. (2021). Performance: An R package for assessment, comparison and testing of statistical models. *Journal of Open Source Software*, 6(60), 3139. <https://doi.org/10.21105/joss.03139>
- Mang, J., Ustjanzew, N., Lesske, I., Schiepe-Tiska, A., & Reiss, K. (2019). *PISA 2015 Skalenhandbuch: Dokumentation der Erhebungsinstrumente [PISA 2015 Scale Manual: Documentation of Survey Instruments]*. Waxmann.
- Matthews, J. S., Ponitz, C. C., & Morrison, F. J. (2009). Early gender differences in self-regulation and academic achievement. *Journal of Educational Psychology*, 101(3), 689–704. <https://doi.org/10.1037/a0014240>
- McDonald, R. P. (1999). *Test theory: A unified treatment*. L. Erlbaum Associates.
- Michalsky, T. (2017). What teachers know and do about assessing students' self-regulated learning. *Teachers College Record*, 119(13), 1–16. <https://doi.org/10.1177/016146811711901313>
- Miele, D. B., Fujita, K., & Scholer, A. A. (2024). The role of metamotivational knowledge in the regulation of motivation. *Motivation Science*, 10(3), 197–209. <https://doi.org/10.1037/mot0000336>
- Muntoni, F., & Retelsdorf, J. (2018). Gender-specific teacher expectations in reading. The role of teachers' gender stereotypes. *Contemporary Educational Psychology*, 54, 212–220. <https://doi.org/10.1016/j.cedpsych.2018.06.012>
- Murray, S. A., Davis, J., Shuler, H. D., Spencer, E. C., & Hinton, A. (2022). Time management for STEMM students during the continuing pandemic. *Trends in Biochemical Sciences*, 47(4), 279–283. <https://doi.org/10.1016/j.tibs.2021.12.010>
- Muschkin, C. G., Malone, P. S., Conduct Problems Prevention Research Group. (2007). Multiple teacher ratings: An evaluation of measurement strategies. *Educational Research and Evaluation*, 13(1), 71–86. <https://doi.org/10.1080/13803610601058215>

- Nakagawa, S., & Schielzeth, H. (2013). A general and simple method for obtaining  $R^2$  from generalized linear mixed-effects models. *Methods in Ecology and Evolution*, 4(2), 133–142. <https://doi.org/10.1111/1/j.2041-210x.2012.00261.x>
- Nickerson, R. S. (1998). Confirmation bias: A ubiquitous phenomenon in many guises. *Review of General Psychology*, 2(2), 175–220. <https://doi.org/10.1037/1089-2680.2.2.175>
- Noor, N., Beram, S., Huat, F. K. C., Gengatharan, K., & Mohamad Rasidi, M. S. (2023). Bias, halo effect and horn effect: A systematic literature review. *International Journal of Academic Research in Business and Social Sciences*, 13(3), 1116–1140. <https://doi.org/10.6007/IJARBS/v13-i3/16733>
- OECD (2019). *OECD future of education and skills 2030. OECD learning compass 2030. A series of concept notes*.
- Onan, E., Biwer, F., Abel, R., Wiradhany, W., & de Bruin, A. (2024). Optimizing self-organized study orders: Combining refutations and metacognitive prompts improves the use of interleaved practice. *Npj Science of Learning*, 9(1), Article 33. <https://doi.org/10.1038/s41539-024-00245-7>
- Panadero, E. (2017). A review of self-regulated learning: Six models and four directions for research. *Frontiers in Psychology*, 8, 422. <https://doi.org/10.3389/fpsyg.2017.00422>
- Panadero, E., Klug, J., & Järvelä, S. (2016). Third wave of measurement in the self-regulated learning field: When measurement and intervention come hand in hand. *Scandinavian Journal of Educational Research*, 60(6), 723–735. <https://doi.org/10.1080/00313831.2015.1066436>
- Paulick, I., Großschädl, J., Harms, U., & Möller, J. (2016). Preservice teachers' professional knowledge and its relation to academic self-concept. *Journal of Teacher Education*, 67(3), 173–182. <https://doi.org/10.1177/0022487116639263>
- Perry, N. E. (2013). Understanding classroom processes that support children's self-regulation of learning. In D. Whitebread, N. Mercer, C. Howe, & A. Tolmie (Eds.), *BJEP Monograph Series II: Part 10 Self-Regulation and Dialogue in Primary Classrooms*. British Psychological Society. <https://doi.org/10.5384/bpsmono.2013.cat1370.4>
- Quinn, D. M. (2020). Experimental evidence on teachers' racial bias in student evaluation: The role of grading scales. *Educational Evaluation and Policy Analysis*, 42(3), 375–392. <https://doi.org/10.3102/0162373720932188>
- R Core Team (2024). *R: A language and environment for statistical computing* [Computer software]. R Foundation for Statistical Computing. <https://www.r-project.org/>
- Radkowsch, A., Sommerhoff, D., Nickl, M., Codreanu, E., Ufer, S., & Seidel, T. (2023). Exploring the diagnostic process of pre-service teachers using a simulation – A latent profile approach. *Teaching and Teacher Education*, 130, Article 104172. <https://doi.org/10.1016/j.tate.2023.104172>
- Retelsdorf, J., Bauer, J., Gebauer, S. K., Kauper, T., & Möller, J. (2014). Erfassung berufsbezogener Selbstkonzepte von angehenden Lehrkräften (ERBSE-L) [Measuring prospective teachers' professional self-concept]. *Diagnostica*, 60(2), 98–110. <https://doi.org/10.1026/0012-1924/a000108>
- Retelsdorf, J., Schwartz, K., & Asbrock, F. (2015). Michael can't read! teachers' gender stereotypes and boys' reading self-concept. *Journal of Educational Psychology*, 107(1), 186–194. <https://doi.org/10.1037/a0037107>
- Richardson, M., Abraham, C., & Bond, R. (2012). Psychological correlates of university students' academic performance: A systematic review and meta-analysis. *Psychological Bulletin*, 138(2), 353–387. <https://doi.org/10.1037/a0026838>
- Rosseel, Y. (2012). Lavan: An R package for structural equation modeling. *Journal of Statistical Software*, 48(2), 1–36. <https://doi.org/10.18637/jss.v048.i02>
- Rost, F. (2018). Der häusliche Arbeitsplatz und die Arbeitsmittel [The home-based workplace and work equipment]. In F. Rost, *Lern- und Arbeitstechniken für das Studium [Learning and study techniques for university]* (pp. 85–107). Springer Fachmedien Wiesbaden. [https://doi.org/10.1007/978-3-658-17626-6\\_5](https://doi.org/10.1007/978-3-658-17626-6_5)
- Sanrey, C., Bressoux, P., Lima, L., & Pansu, P. (2021). A new method for studying the halo effect in teachers' judgement and its antecedents: Bringing out the role of certainty. *British Journal of Educational Psychology*, 91(2), 658–675. <https://doi.org/10.1111/bjep.12385>
- Schmidt, F. T. C., Kaiser, A., & Retelsdorf, J. (2023). Halo effects in grading: An experimental approach. *Educational Psychology*, 43(2–3), 246–262. <https://doi.org/10.1080/01443410.2023.2194593>
- Schraw, G., Crippen, K. J., & Hartley, K. (2006). Promoting self-regulation in science education: Metacognition as part of a broader perspective on learning. *Research in Science Education*, 36(1–2), 111–139. <https://doi.org/10.1007/s11165-005-3917-8>
- Sneyers, E., Vanhoof, J., & Mahieu, P. (2019). The impact of primary school teachers' expectations of pupils, parents and teachers on teacher track recommendations. *Educational Studies*, 55(3), 327–345. <https://doi.org/10.1080/00131946.2018.1562925>
- Sneyers, E., Vanhoof, J., & Mahieu, P. (2020). Bias in primary school teachers' expectations of students? A study of general and specific bias towards SES, ethnicity and gender. *Studia Paedagogica*, 25(2), 71–96. <https://doi.org/10.5817/SP2020-2-4>

- Spirgel, A. S., & Delaney, P. F. (2016). Does writing summaries improve memory for text? *Educational Psychology Review*, 28(1), 171–196. <https://doi.org/10.1007/s10648-014-9290-2>
- Sprietsma, M. (2013). Discrimination in grading: Experimental evidence from primary school teachers. *Empirical Economics*, 45(1), 523–538. <https://doi.org/10.1007/s00181-012-0609-x>
- Størksen, I., Ellingsen, I. T., Wanless, S. B., & McClelland, M. M. (2015). The influence of parental socioeconomic background and gender on self-regulation among 5-year-old children in Norway. *Early Education and Development*, 26(5–6), 663–684. <https://doi.org/10.1080/10409289.2014.932238>
- Südkamp, A., Kaiser, J., & Möller, J. (2012). Accuracy of teachers' judgments of students' academic achievement: A meta-analysis. *Journal of Educational Psychology*, 104(3), 743–762. <https://doi.org/10.1037/a0027627>
- Tan, C. Y. (2024). Socioeconomic status and student learning: Insights from an umbrella review. *Educational Psychology Review*, 36, 100. <https://doi.org/10.1007/s10648-024-09929-3>
- Tannert, S., Karlen, Y., Ader, E., Hertel, S., & Glogger-Frey, I. (2026). Teachers' assessment competence in self-regulated learning – presence of non-diagnostic cues diminishes judgment accuracy. *Metacognition and Learning*, 21(1), 14. <https://doi.org/10.1007/s11409-026-09455-3>
- Taxis, S. S., Stralla, M., Estner, C., Herzmann, P., & Seufert, T. (2017). Clever lernen [smart learning]. In C. Gräsel & K. Trempler (Eds.), *Entwicklung von Professionalität pädagogischen Personals [development of professional competence among educational staff]* (pp. 237–259). Springer Fachmedien Wiesbaden. [https://doi.org/10.1007/978-3-658-07274-2\\_13](https://doi.org/10.1007/978-3-658-07274-2_13)
- Tobisch, A., & Dresel, M. (2017). Negatively or positively biased? Dependencies of teachers' judgments and expectations based on students' ethnic and social backgrounds. *Social Psychology of Education*, 20(4), 731–752. <https://doi.org/10.1007/s11218-017-9392-z>
- Urhahne, D., Timm, O., Zhu, M., & Tang, M. (2013). Sind unterschätzte Schüler weniger leistungsmotiviert als überschätzte Schüler? [Are underestimated students less achievement motivated than overestimated students?]. *Zeitschrift für Entwicklungspsychologie und Pädagogische Psychologie [Journal of Developmental Psychology and Educational Psychology]*, 45(1), 34–43. <https://doi.org/10.1026/0049-8637/a000079>
- Urhahne, D., & Wijnia, L. (2021). A review on the accuracy of teacher judgments. *Educational Research Review*, 32, Article 100374. <https://doi.org/10.1016/j.edurev.2020.100374>
- Van den Broeck, L., De Laet, T., Dujardin, R., Tuybaerts, S., & Langie, G. (2024). Unveiling the competencies at the core of lifelong learning: A systematic literature review. *Educational Research Review*, 45, Article 100646. <https://doi.org/10.1016/j.edurev.2024.100646>
- van Leeuwen, A., Teasley, S. D., & Wise, A. F. (2022). Teacher and student facing learning analytics. In C. Lang, G. Siemens, A. F. Wise, D. Gašević & A. Merceron (Eds.), *The Handbook of Learning Analytics* (2nd ed., pp. 130–140). SOLAR. <https://doi.org/10.18608/hla22.013>
- Vanlommel, K., Van Gasse, R., Vanhoof, J., & Van Petegem, P. (2017). Teachers' decision-making: Data based or intuition driven? *International Journal of Educational Research*, 83, 75–83. <https://doi.org/10.1016/j.ijer.2017.02.013>
- Verbree, A.-R., Hornstra, L., Maas, L., & Wijngaards-de Meij, L. (2023). Conscientiousness as a predictor of the gender gap in academic achievement. *Research in Higher Education*, 64(3), 451–472. <https://doi.org/10.1007/s11162-022-09716-5>
- Waldeyer, J., Fleischer, J., Wirth, J., & Leutner, D. (2020). Validating the Resource-Management Inventory (ReMI): Testing measurement invariance and predicting academic achievement in a sample of first-year university students. *European Journal of Psychological Assessment*, 36(5), 777–786. <https://doi.org/10.1027/1015-5759/a000557>
- Wargocki, P., Porras-Salazar, J. A., Contreras-Espinoza, S., & Bahnfleth, W. (2020). The relationships between classroom air quality and children's performance in school. *Building and Environment*, 173, Article 106749. <https://doi.org/10.1016/j.buildenv.2020.106749>
- Wesarg-Menzel, C., Ebbes, R., Hensums, M., Wagemaker, E., Zaharieva, M. S., Staaks, J. P. C., van den Akker, A. L., Visser, I., Hoeve, M., Brummelman, E., Dekkers, T. J., Schuitema, J. A., Larsen, H., Colomnesi, C., Jansen, B. R. J., Overbeek, G., Huizenga, H. M., & Wiers, R. W. (2023). Development and socialization of self-regulation from infancy to adolescence: A meta-review differentiating between self-regulatory abilities, goals, and motivation. *Developmental Review*, 69, Article 101090. <https://doi.org/10.1016/j.dr.2023.101090>
- Wetzel, C. G., Wilson, T. D., & Kort, J. (1981). The halo effect revisited: Forewarned is not forearmed. *Journal of Experimental Social Psychology*, 17(4), 427–439. [https://doi.org/10.1016/0022-1031\(81\)90049-4](https://doi.org/10.1016/0022-1031(81)90049-4)
- Won, S., Lee, S. Y., & Bong, M. (2017). Social persuasions by teachers as a source of student self-efficacy: The moderating role of perceived teacher credibility. *Psychology in the Schools*, 54(5), 532–547. <https://doi.org/10.1002/pits.22009>

- Yeung, A. S., Craven, R. G., & Kaur, G. (2014). Teachers' self-concept and valuing of learning: Relations with teaching approaches and beliefs about students. *Asia-Pacific Journal of Teacher Education*, 42(3), 305–320. <https://doi.org/10.1080/1359866X.2014.905670>
- Zhou, J., & Urhahne, D. (2013). Teacher judgment, student motivation, and the mediating effect of attributions. *European Journal of Psychology of Education*, 28(2), 275–295. <https://doi.org/10.1007/s10212-012-0114-9>
- Zhu, M., Urhahne, D., & Rubie-Davies, C. M. (2018). The longitudinal effects of teacher judgement and different teacher treatment on students' academic outcomes. *Educational Psychology*, 38(5), 648–668. <https://doi.org/10.1080/01443410.2017.1412399>
- Zimmerman, B. J. (2000). Attaining self-regulation: A social cognitive perspective. In M. Boekaerts, P. R. Pintrich, & M. Zeidner (Eds.), *Handbook of self-regulation* (pp. 13–39). Academic.
- Zimmerman, B. J. (2002). Becoming a self-regulated learner: An overview. *Theory into Practice*, 41(2), 64–70. [https://doi.org/10.1207/s15430421tip4102\\_2](https://doi.org/10.1207/s15430421tip4102_2)
- Zuur, A. F., Ieno, E. N., & Elphick, C. S. (2010). A protocol for data exploration to avoid common statistical problems. *Methods in Ecology and Evolution*, 1(1), 3–14. <https://doi.org/10.1111/j.2041-210X.2009.00001.x>

**Publisher's note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.